

Quantum Machine Learning for CIFAR-10 Image Classification:

A Comparative Study of Hybrid Quantum-Classical Architectures,
Multi-Scale Fusion, Learnable Quadvolution, and Encoding Strategies

Bachelor's Thesis Project — Quantum Machine Learning

Mantavya

Department of Computer Science and Engineering
Indraprastha Institute of Information Technology, Delhi (IIIT-Delhi)

July 22, 2026

Abstract

This report consolidates six independent experimental notebooks developed over the course of a Bachelor’s Thesis Project (BTP) investigating hybrid quantum–classical pipelines for CIFAR-10 image classification. The experiments span: (i) a hybrid CNN-feature-extractor with a residual Quantum Convolutional Neural Network (QCNN) classifier benchmarked across ten quantum data-encoding strategies; (ii) a Feature-Pyramid-Network-inspired multi-scale extension of the same architecture (FPN-QCNN) together with a diagnosed and partially-repaired follow-up run; (iii) a fully learnable, patch-based quantum convolutional (“quanvolutional”) feature extractor trained end-to-end on classically-derived intensity/edge/texture channels; (iv) a follow-up variant of the same learnable quanvolutional pipeline that instead consumes raw RGB channels directly and adds a second, trainable “fusion” quanvolution layer mixing the three colour channels before classification (a two-level quantum feature pyramid); (v) a large-scale benchmark of eighteen quantum data-encoding strategies — including a novel application of Pauli Correlation Encoding (PCE) — combined with quantum-kernel support vector machines and variational quantum classifiers (VQCs); and (vi) a completed follow-up to the single-scale hybrid pipeline that finishes the full ten-encoding benchmark, sweeps register size from 2 to 8 qubits, and ablates the quantum circuit against its classical residual/skip scaffolding. Across all experiments, the best-performing configuration is the single-scale hybrid CNN+QCNN pipeline with an AQHE-DUALENC encoding, PCA→shrinkage-LDA dimensionality reduction, and a ResNet-style quantum residual connection, reaching 62.0% test accuracy on a balanced 5,000-image, 10-class subset of CIFAR-10 (chance level 10%). Multi-scale (FPN-style) quantum fusion and both learnable-quanvolution variants were found to under-perform this single-scale baseline at the qubit budgets and sample sizes tractable on classical simulators; notably, adding a second (fusion) quanvolution layer and switching to raw RGB channels made the quanvolutional pipeline *worse*, not better (21.7% vs. 26.0% test accuracy), underscoring that naively adding depth or richer raw inputs to an already information-starved quantum feature path compounds the bottleneck rather than relieving it. A follow-up run of the single-scale pipeline (Experiment VI) that completed the full ten-encoding benchmark and added a quantum-vs-classical ablation found that removing the classical skip connection costs far more accuracy (−0.205) than removing the quantum residual re-upload (−0.013), and also surfaced internal inconsistencies (the same configuration scoring differently across sections of the same notebook run) that are reported transparently rather than resolved. The report documents in detail the diagnosed causes of the multi-scale and quanvolution under-performance (information starvation, shallow/over-deep fusion, and aggressive learning-rate decay) together with the partial fixes attempted. The large-scale encoding benchmark shows that no single-qubit angle-style encoding dominates and that the PCE encoding, despite favourable theoretical gate and capacity scaling, underperforms the best LDA-based encodings on this task and dataset size.

Contents

1	Introduction	7
1.1	Motivation	7
1.2	Scope of this report	7
2	Background and Related Work	9
2.1	The CIFAR-10 dataset	9
2.2	Quantum Convolutional Neural Networks (QCNN)	9
2.3	Feature Pyramid Networks	9
2.4	Quanvolutional neural networks	9
2.5	Quantum data encodings and quantum kernels	10
2.6	Pauli Correlation Encoding	10
2.7	Dimensionality reduction: PCA and shrinkage LDA	10
3	Experiment I — Hybrid CNN + Residual QCNN (v6): Single-Scale Benchmark and First FPN-QCNN Attempt	11
3.1	Overview	11
3.2	Dataset and preprocessing	11
3.3	Architecture	12
3.3.1	QCNN building blocks	12
3.3.2	ResNet-style quantum residual connection	12
3.3.3	Ten quantum data encodings	12
3.4	Training procedure	13
3.4.1	v3–v6 changelog (summary)	13
3.5	Results: single-scale benchmark	14
3.6	The Feature-Pyramid multi-scale extension (FPN-QCNN), first attempt	15
3.6.1	Motivation	15
3.6.2	Mapping classical FPN onto the quantum pipeline	15
3.6.3	Configuration	15
3.6.4	Result	15
4	Experiment II — Refined Hybrid CNN + Residual QCNN (v6.1): Diagnosing and Repairing the FPN-QCNN Under-Performance	16
4.1	Overview	16
4.2	Single-scale re-benchmark with a longer CNN training budget	16
4.3	Root-cause diagnosis of the FPN-QCNN under-performance	17
4.4	Fixes applied and re-run configuration	17
4.5	Result: v6.1 FPN-QCNN	18

5	Experiment III — Learnable Quanvolutional Feature Extraction	19
5.1	Overview	19
5.2	Dataset and preprocessing	19
5.3	Architecture	19
5.3.1	Learnable quanvolution layer	19
5.3.2	Downstream classifier	20
5.3.3	Joint end-to-end optimization	20
5.4	Computational cost	20
5.5	Results	20
5.6	Suggested next steps (as recorded in the notebook)	22
6	Experiment IV — Learnable Quanvolution with Raw RGB Channels and a Second Fusion Layer (Two-Level Quantum Feature Pyramid)	24
6.1	Overview	24
6.2	Dataset and preprocessing	24
6.3	Architecture	25
6.3.1	Level-1 quanvolution: per-channel, patch-shared circuits	25
6.3.2	Level-2 quanvolution: the fusion layer	25
6.3.3	Downstream classifier	25
6.3.4	Joint end-to-end optimization	25
6.4	Computational cost	25
6.5	Results	26
6.6	Suggested next steps (as recorded in the notebook)	28
7	Experiment V — Unified Benchmark of 18 Quantum Encodings with Pauli Correlation Encoding	29
7.1	Overview	29
7.2	Dataset and dual preprocessing pipelines	29
7.3	The eighteen quantum encodings	30
7.3.1	Seventeen standard encodings	30
7.3.2	Pauli Correlation Encoding (PCE)	31
7.4	Classifiers: quantum-kernel SVM and VQC	32
7.5	Main benchmark: all 18 encodings	32
7.6	Qubit-count sensitivity sweep	35
7.7	LDA dimensionality sensitivity	36
7.8	Full VQC training run	37
7.9	Refining PCE: reduced feature count and shallower ansatz (PCE v2)	38
7.10	Summary of observations (Experiment V)	38
8	Experiment VI — Hybrid CNN + Residual QCNN (v7): Full 10-Encoding Benchmark, Qubit-Count Sweep, and Quantum-vs-Classical Ablation	40
8.1	Overview	40
8.1.1	v7 changelog	40
8.2	Dataset and preprocessing	41
8.3	Full 10-encoding benchmark	41
8.4	Qubit-count sweep	43
8.5	Quantum-vs-classical ablation	43
8.6	Widened FPN-QCNN encoding sweep	44
8.7	QCNN-vs-kernel-SVM rank correlation	45
8.8	Assumptions, caveats, and notebook-state inconsistencies	45
8.9	Summary of observations (Experiment VI)	46

9	Cross-Experiment Discussion	47
9.1	Consolidated results	47
9.2	Recurring themes across experiments	48
10	Conclusion and Future Work	50
10.1	Conclusion	50
10.2	Future work	50

List of Figures

3.1	Training and validation accuracy curves for the three benchmarked encodings (AQHE-DualEnc, FRQI, IQP), Experiment I single-scale hybrid CNN+QCNN pipeline. Dotted grey line marks chance level (0.10).	14
4.1	Training and validation accuracy curves, Experiment II single-scale re-benchmark (longer CNN training budget). Dotted grey line marks chance level (0.10).	17
5.1	Training and validation accuracy curve, Experiment III learnable quanvolution + QCNN classifier. Dotted grey line marks chance level (0.10).	21
5.2	Qualitative check: raw (pre-quantum, classical) intensity/edge/texture patch grids for four representative test images, used to sanity-check what information the trainable quantum channels are actually operating on.	22
6.1	Training and validation accuracy curve, Experiment IV (RGB channels, two-level fusion quanvolution + QCNN classifier). Dotted grey line marks chance level (0.10). Training halts at epoch 10 due to early stopping.	26
6.2	Qualitative check: raw (pre-quantum, classical) R, G, B patch grids for four representative test images, used to sanity-check what information the trainable quantum channels are actually operating on in this RGB variant.	27
7.1	Dual-pipeline architecture, Experiment V. Pipeline A (top) feeds 17 standard encodings via PCA-50→LDA-8; Pipeline B (bottom) feeds Pauli Correlation Encoding via PCA-64 directly.	30
7.2	First two LDA discriminant components of the CIFAR-10 training subsample (Pipeline A), colour-coded by class.	30
7.3	PCE encoding capacity $m_{\max}(n, k) = \binom{n}{k} \times 3^k$ as a function of qubit count n , for correlation orders $k \in \{1, 2, 3\}$. The chosen operating point ($n = 8, k = 2$) lies inside the feasible region for the target $m = 64$ features.	32
7.4	Summary heat map of quantum-kernel SVM test accuracy across all 18 encodings on CIFAR-10 (10 classes).	34
7.5	Comparative bar chart of quantum-kernel SVM test accuracy across all benchmarked encodings.	35
7.6	Encoding-family radar/spider chart comparing accuracy across encoding groups.	35
7.7	Test accuracy as a function of qubit count (= LDA dimensionality), swept from 2 to 8 qubits, for all encodings except PCE. Dashed grey line marks chance level (0.10).	36
7.8	Test accuracy vs. LDA dimensionality (= qubit count) for four representative encodings; each point refits LDA from scratch. LDA is hard-capped at $K - 1 = 9$ components for $K = 10$ classes.	37
7.9	VQC (2-layer, FRQI encoding) training curve on CIFAR-10 (10 classes), 100 epochs. Solid = train, dashed = validation, final point = held-out test.	37

8.1	Training and validation accuracy curves for all ten benchmarked encodings, Experiment VI full sweep. Dotted grey line marks chance level (0.10).	42
8.2	Test accuracy by quantum encoding, Experiment VI full 10-encoding sweep. Bar colour highlights the best-performing encoding (Pi4Angle).	42
8.3	Test accuracy vs. register size (qubit count), Experiment VI qubit-count sweep, Pi4Angle encoding.	43
8.4	Ablation bar chart, Experiment VI: full model vs. removing the residual re-upload, removing the classical skip connection, and removing both (quantum-circuit-only). Red bar marks the quantum-only configuration.	44

List of Tables

3.1	Quantum data-encoding circuits benchmarked in Experiment I.	13
3.2	Experiment I single-scale hybrid CNN+QCNN test accuracy (5,000 images, chance = 10%).	14
4.1	Experiment II single-scale hybrid CNN+QCNN test accuracy, longer CNN training budget (compare with Table 3.2).	16
4.2	FPN-QCNN test accuracy before and after the Sec. 4.3 fixes.	18
5.1	Experiment III learnable quanvolution + QCNN classifier results (1,500 images, chance = 10%).	20
6.1	Experiment IV (RGB channels, two-level fusion quanvolution) results, 1,500 images, chance = 10%, compared against Experiment III (intensity/edge/texture, single layer).	26
7.1	The seventeen standard quantum encodings benchmarked in Experiment V (grouped by family).	31
7.2	Experiment V main benchmark: quantum-kernel SVM test accuracy, all 18 encodings.	33
7.3	PCE ablation: baseline (v1, PCA-64, 8-layer HEA) vs. refined (v2, PCA-24 by F-score, 2-layer dual-axis HEA, tuned C), against the best LDA-based encoding.	38
8.1	Experiment VI full 10-encoding hybrid CNN+QCNN benchmark, test accuracy (5,000 images, chance = 10%).	41
8.2	Experiment VI qubit-count sweep, Pi4Angle encoding, test accuracy vs. register size.	43
8.3	Experiment VI ablation: quantum circuit vs. classical scaffolding, Pi4Angle encoding.	44
9.1	Cross-experiment summary of best-reported test accuracy.	47

Chapter 1

Introduction

1.1 Motivation

Quantum Machine Learning (QML) explores whether parameterized quantum circuits, used either as trainable classifiers or as feature-space kernels, can offer any representational or sample-efficiency advantage over purely classical models on practical learning tasks [5, 6]. Image classification on natural-image benchmarks such as CIFAR-10 [11] is considerably harder than the toy datasets (MNIST, Iris, synthetic blobs) on which most QML architectures are first demonstrated, because (a) natural images require genuinely spatial/textural feature extraction before any low-dimensional embedding can be handed to a small quantum register, and (b) classical simulation of quantum circuits scales exponentially in qubit count, which caps the amount of raw information that can realistically be fed into a simulated circuit.

This Bachelor’s Thesis Project (BTP) investigates several architectural strategies for closing this gap: hybridizing classical convolutional feature extraction with a Quantum Convolutional Neural Network (QCNN) classifier [1]; borrowing the multi-scale Feature Pyramid Network (FPN) design [2] to give the quantum circuit access to features from multiple spatial resolutions; making the quantum feature-extraction stage itself learnable via a quanvolutional layer [3]; and benchmarking a wide range of classical-to-quantum data-encoding strategies, including quantum kernel methods [4, 5] and a recently proposed qubit-efficient encoding, Pauli Correlation Encoding (PCE) [7], repurposed here from its original combinatorial-optimization setting into a data-encoding scheme for supervised classification.

1.2 Scope of this report

This report compiles the methodology, architecture, assumptions, datasets, procedures, and quantitative/qualitative results recorded across four Jupyter notebooks produced during the BTP:

- (I) `QCNN_CIFAR10_Hybrid_v6.ipynb` — Hybrid CNN + Residual QCNN, single-scale benchmark across ten encodings, plus a first Feature-Pyramid multi-scale QCNN (FPN-QCNN) variant (Chapter 3).
- (II) `QCNN_CIFAR10_Hybrid_v6_1.ipynb` — A refined re-run of the same pipeline with an enlarged CNN training budget and a corrected, information-rich FPN-QCNN configuration, together with a detailed root-cause diagnosis of why the multi-scale variant under-performs the single-scale baseline (Chapter 4).
- (III) `QCNN_CIFAR10_LearnableQuanvolution_v1.ipynb` — A fully learnable, patch-based quantum convolutional (quanvolutional) feature extractor trained jointly with a QCNN classifier head (Chapter 5).

- (IV) `QCNN_CIFAR10_LearnableQuanvolution_v1_-_Copy.ipynb` — A follow-up run of the same learnable quanvolutional pipeline, switched to raw RGB input channels and extended with a second, trainable “fusion” quanvolution layer that mixes the three colour channels before classification, i.e. a two-level quantum feature pyramid (Chapter 6).
- (V) `QML_CIFAR10_v3_PCE_fixed.ipynb` — A unified benchmark of eighteen quantum data-encoding strategies (seventeen standard encodings plus Pauli Correlation Encoding) combined with quantum-kernel SVMs and variational quantum classifiers, with qubit-count and LDA-dimensionality sensitivity sweeps (Chapter 7).
- (VI) `QCNN_CIFAR10_Hybrid_v7.ipynb` — A completed follow-up to Experiments I–II that trains and tests all ten single-scale encodings to completion, sweeps register size (qubit count) from 2 to 8 qubits, and runs a quantum-vs-classical ablation isolating the residual re-upload and classical skip connection’s individual contributions (Chapter 8).

Each experiment is presented in its own chapter with dedicated subsections for architecture, dataset and preprocessing, training procedure, assumptions/limitations, and results (tables and figures reproduced from the notebooks’ own outputs). Chapter 9 draws cross-experiment comparisons, and Chapter 10 concludes with directions for future work.

Chapter 2

Background and Related Work

2.1 The CIFAR-10 dataset

CIFAR-10 [11] consists of 60,000 32×32 RGB natural images evenly split across 10 mutually exclusive classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck). All four experiments in this report use class-balanced subsamples of CIFAR-10 (ranging from 600 to 5,000 images) rather than the full dataset, because the per-sample cost of simulating parameterized quantum circuits on classical hardware (either via state-vector simulation or quantum-kernel evaluation) makes the full 60,000-image dataset computationally intractable during iterative BTP development.

2.2 Quantum Convolutional Neural Networks (QCNN)

The QCNN architecture of [1] alternates parameterized two-qubit *convolution* blocks (translationally-invariant unitaries applied to neighbouring qubit pairs) with *pooling* blocks that conditionally discard qubits, mirroring the convolution/pooling structure of classical CNNs while respecting the no-cloning constraints of quantum circuits. All experiments in this report reuse a common conv/pool block design (Sec. 3.3.1) built from parameterized $U3/\text{CRot}$ gates (convolution) and controlled-rotation gates (pooling).

2.3 Feature Pyramid Networks

Feature Pyramid Networks (FPN) [2] address the classical object-detection problem of recognizing objects at multiple scales by building a top-down pathway with lateral connections that fuses coarse, semantically strong features with fine, spatially precise features. Experiments I and II (Chapters 3–4) adapt the *spirit* of FPN’s top-down multi-scale fusion, and of anchor-style fixed-region pooling, into a multi-register quantum circuit, using entangling gates as the quantum stand-in for FPN’s additive lateral fusion.

2.4 Quanvolutional neural networks

[3] proposed replacing (or augmenting) classical convolutional filters with small, fixed random quantum circuits applied patch-wise across an image (“quandvolution”), producing a transformed feature map that is then consumed by a classical or quantum classifier. Experiment III (Chapter 5) extends this idea by making all quanvolutional circuit parameters *trainable* and optimizing them jointly with the downstream classifier via backpropagation-based gradient descent. Experiment IV (Chapter 6) follows up directly on Experiment III, stacking a second, trainable “fusion”

quantum convolution layer on top of the first so that channel-level features can be mixed before pooling into the classifier — the quantum analogue of stacking a second convolutional layer in a classical CNN.

2.5 Quantum data encodings and quantum kernels

Any classical feature vector must be mapped into a quantum state before a quantum circuit can process it; the choice of this encoding circuit critically affects both circuit expressivity and optimization landscape [5]. [4] showed that a data-encoding feature map can equivalently be used to define a quantum kernel for use inside a classical support vector machine (SVM), decoupling the (classically hard to simulate) feature map from the (classically tractable) convex optimization of the SVM. Experiment V (Chapter 7) benchmarks eighteen such encodings, both directly through quantum-kernel SVMs and, for a subset, through gradient-trained variational quantum classifiers (VQCs) [6].

2.6 Pauli Correlation Encoding

Pauli Correlation Encoding (PCE) was introduced by [7] as a qubit-efficient variational scheme for combinatorial optimization problems (e.g. Max-Cut), in which $m = O(n^k)$ binary variables are encoded into the signs of k -body Pauli-correlator expectation values measured on only n qubits, giving a capacity $m_{\max}(n, k) = \binom{n}{k} \times 3^k$. Experiment V repurposes this qubit-efficient encoding capacity as a *data encoding* for supervised image classification: rather than encoding m optimization variables, it encodes m continuous-valued classical features (from a PCA projection of CIFAR-10) into a hardware-efficient-ansatz (HEA) circuit on $n = 8$ qubits, and defines a quantum kernel from the resulting statevector overlaps.

2.7 Dimensionality reduction: PCA and shrinkage LDA

All four experiments reduce high-dimensional pixel or embedding data down to the small number of dimensions (≤ 9 , one per available qubit typically) that a classically-simulated quantum circuit can absorb, using Principal Component Analysis (PCA) followed, where labels are used, by Linear Discriminant Analysis (LDA). Because LDA is capped at $K - 1 = 9$ components for $K = 10$ classes and its class-covariance estimate is unstable when fit on a few thousand samples, several notebooks (Experiments I and II) adopt Ledoit–Wolf shrinkage-regularized LDA [12] in place of vanilla LDA.

Chapter 3

Experiment I — Hybrid CNN + Residual QCNN (v6): Single-Scale Benchmark and First FPN-QCNN Attempt

3.1 Overview

This experiment (notebook `QCNN_CIFAR10_Hybrid_v6.ipynb`) is the sixth iteration (v2→v6) of the BTP’s main hybrid pipeline. Earlier iterations (v2–v5, documented as in-notebook changelogs and summarized in Sec. 3.4.1) diagnosed and fixed a sequence of failure modes — near-chance accuracy from a raw-pixel PCA front end, severe overfitting, and an optimizer plateau — before arriving at the architecture described below. Section 3.6 then extends the pipeline with a Feature-Pyramid-style multi-scale quantum fusion architecture (FPN-QCNN) and reports its first (under-performing) result.

3.2 Dataset and preprocessing

- **Dataset:** CIFAR-10, class-balanced subsample of 500 images/class (5,000 images total), drawn from the full 60,000-image pool.
- **Split:** 80% train+val / 20% test, with a further 10% of the train+val portion held out for validation (3,600 train / 400 val / 1,000 test).
- **Classical CNN feature extractor:** A convolutional network (trained from scratch on the same 5,000-image subsample) reduces each $32 \times 32 \times 3$ image to a 128-dimensional embedding. Training uses random-crop and horizontal-flip augmentation, Adam with weight decay 5×10^{-4} , dropout 0.3 before the final fully-connected layer, and best-validation-accuracy checkpointing with early stopping (patience 6). This CNN reached a best validation accuracy of 63.7% on its own 10-class softmax head after 25 epochs.
- **Dimensionality reduction:** `StandardScaler` → `PCA(16)` on the 128-d CNN embedding (retaining 96.0% of variance) → shrinkage LDA (solver `eigen`, Ledoit–Wolf `shrinkage="auto"`) down to 8 discriminant components → `MinMaxScaler` to $[0, \pi]$ for angle-encoding compatibility.
- **Classical residual/skip features:** the first 8 (standardized) dimensions of the CNN embedding are retained separately and concatenated directly onto the final classifier input, bypassing the quantum circuit entirely for these dimensions — the classical half of a ResNet-style shortcut.

3.3 Architecture

3.3.1 QCNN building blocks

Every quantum classifier in this report (Experiments I–III) is built from the same three primitives:

- **Convolution block** (9 parameters): two single-qubit $U3(\theta_1, \theta_2, \theta_3)$ rotations followed by a two-qubit $\text{CRot}(\theta_7, \theta_8, \theta_9)$ entangling gate, applied to every (cyclically) neighbouring qubit pair.
- **Pooling block** (2 parameters): a controlled- R_Z and a controlled- R_X (sandwiching a Pauli- X to flip control polarity) that conditionally rotates a “target” qubit based on a “control” qubit, after which the control qubit is discarded from the active register — the quantum analogue of a $2\times$ pooling step.
- **Classifier head**: a `StronglyEntanglingLayers` ansatz (2 layers) over the surviving qubits, followed by multi-basis readout (Pauli- Z , X , and Y expectation values on each surviving qubit).

An 8-qubit input register is compressed via repeated conv/pool layers down to 4 surviving qubits (`CONV_DEPTH=2` convolution repetitions per layer).

3.3.2 ResNet-style quantum residual connection

At the start of every conv/pool layer, a scaled copy of the *original* encoded input is re-uploaded onto the still-active wires via extra R_Y rotations, weighted by `RESIDUAL_REUPLOAD_SCALE=0.3`. This is the closest quantum-circuit analogue of a ResNet identity shortcut $y = F(x) + x$: it prevents the original signal from being fully “washed out” after several rounds of pooling, mirroring the purpose of residual connections in deep classical networks [10].

3.3.3 Ten quantum data encodings

Table 3.1 lists the ten encoding circuits benchmarked in this experiment. All encodings act on the 8-dimensional PCA→LDA feature vector, chunked into one 8-qubit register (`SHARE_WEIGHTS_ACROSS_CHUNKS=True`).

Table 3.1: Quantum data-encoding circuits benchmarked in Experiment I.

Encoding	Circuit description
Pi4Angle	Hadamard superposition + $R_Y(2x_i)$ per qubit.
EntangledAngle	Hadamard + $R_Y(x_i)$ per qubit, followed by a linear CNOT chain.
DenseAngle	Hadamard + $R_Z(x_i^2)$, plus cross-term $R_Z(x_i x_j)$ via CNOT-sandwiched rotations for nearby qubit pairs.
FRQI	Hadamard superposition + circular controlled- $R_Y(2x_i)$ gates (Flexible Representation of Quantum Images) [8].
IQP	Two layers of Hadamard + $R_Z(x_i)$ + all-pairs $R_Z(x_i x_j)$ cross terms (Instantaneous Quantum Polynomial-time circuit) [4].
QHE	Hadamard + PhaseShift ($2\pi x_i$) + inverse Quantum Fourier Transform (Quantum Helix Encoding).
Hamiltonian	Hadamard + first-order Trotterized evolution under an Ising-like Hamiltonian (R_Z single-qubit + R_X two-qubit terms), 2 Trotter steps.
AQHE-LogHelix	Hadamard + logarithmically-warped PhaseShift + inverse QFT.
AQHE-FibSpiral	Golden-angle (ϕ -based) spiral phase placement via R_Y + PhaseShift + inverse QFT.
AQHE-DualEnc	Combines an R_Y amplitude term with a logarithmically-warped phase term, followed by inverse QFT.

3.4 Training procedure

Adam-style optimization (`qml.AdamOptimizer`) with:

- Learning rate 0.02 with reduce-on-plateau decay (factor 0.5, patience 2 epochs);
- L2 regularization on the linear classifier head/FC layers ($\lambda = 0.006$) and separately, more weakly, on the QCNN conv/pool weights ($\lambda_{\text{circuit}} = 0.0005$);
- Batch size 64, up to 30 epochs, early stopping with patience 8 epochs *measured from the last learning-rate decay* (not from the last validation improvement);
- Best-validation-accuracy checkpoint restoration at the end of training;
- `lightning.qubit` simulator backend with adjoint differentiation. A circuit returning raw `qml.probs()` was found to silently disable adjoint differentiation (falling back to $O(\text{parameters})$ -cost parameter-shift rule); switching the circuit to return only expectation values restored adjoint differentiation and was the single largest wall-clock speed fix applied across the BTP.

3.4.1 v3–v6 changelog (summary)

v2→v3: Replaced a raw-pixel PCA(8) front end (which discarded almost all spatial/textural signal and left several encodings scoring *below* chance) with the CNN+PCA+LDA pipeline above, and increased the dataset from 400 to 5,000 images.

v3→v4: Diagnosed severe overfitting (train accuracy jumping to 85–96% within 1–2 epochs while validation accuracy plateaued/oscillated). Fixed via a lower learning rate ($0.05 \rightarrow 0.008$), broadened/strengthened L2 regularization (including, newly, the QCNN conv/pool

weights), early stopping, one fewer classifier-head layer, shrinkage LDA, and CNN-side augmentation/weight-decay/dropout/checkpointing.

v4→v5: Diagnosed an over-correction: training now plateaued completely after epoch 2–3 (an optimizer stall, not overfitting, since train/val tracked closely). Fixed via a restored learning rate ($0.008 \rightarrow 0.02$) combined with reduce-LR-on-plateau (rather than a single fixed rate), lowered L2 coefficients, a larger quantum-FC-layer initialization scale, and a raised epoch ceiling ($15 \rightarrow 30$).

v5→v6: Adds the Feature-Pyramid multi-scale extension described in Sec. 3.6 on top of the now-stable v5 single-scale pipeline.

3.5 Results: single-scale benchmark

Owing to the wall-clock cost of quantum-circuit training under classical simulation (each encoding took roughly 2–6 hours), only the top three encodings from the v2 prior ranking were run to completion on the full 5,000-image dataset. Table 3.2 reports held-out test accuracy (never used for model selection).

Table 3.2: Experiment I single-scale hybrid CNN+QCNN test accuracy (5,000 images, chance = 10%).

Encoding	Test accuracy
AQHE-DualEnc	0.620
FRQI	0.594
IQP	0.580

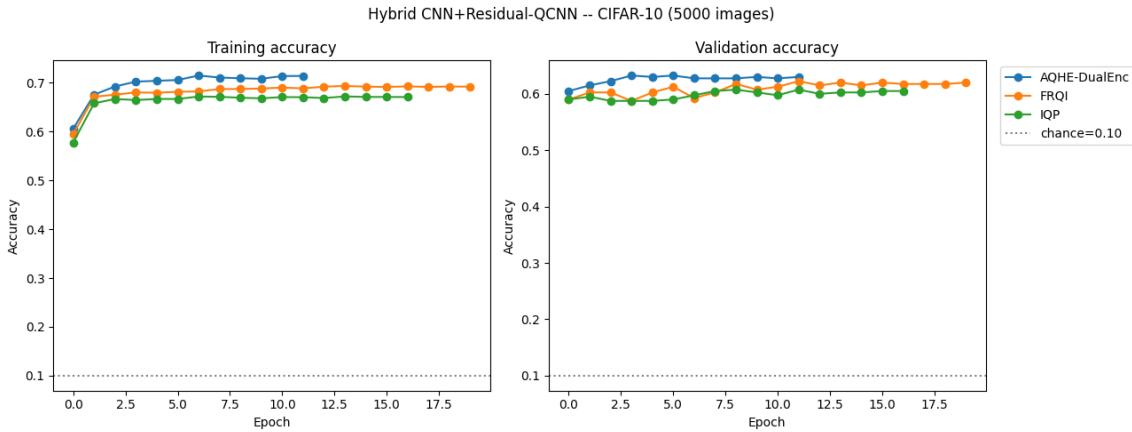


Figure 3.1: Training and validation accuracy curves for the three benchmarked encodings (AQHE-DualEnc, FRQI, IQP), Experiment I single-scale hybrid CNN+QCNN pipeline. Dotted grey line marks chance level (0.10).

3.6 The Feature-Pyramid multi-scale extension (FPN-QCNN), first attempt

3.6.1 Motivation

Every version up to v5 collapses the CNN backbone to a single 128-d penultimate embedding taken after the most-downsampled residual block, which is dominated by global/semantic content and has already discarded fine spatial detail present in earlier layers. Since accuracy plateaued at a similar ceiling across v3–v5 despite fixing purely optimization-related issues, this section tests whether feeding the QCNN *multi-scale* features (local edges/texture, mid-level parts, and global/semantic content, simultaneously) can raise that ceiling.

3.6.2 Mapping classical FPN onto the quantum pipeline

- **Top-down lateral fusion** transfers directly: a small `FPNSmallResNet` backbone produces three feature maps C_1, C_2, C_3 at decreasing spatial resolution, fused top-down via 1×1 lateral convolutions and upsampling, exactly as in [2].
- **Anchors** are re-purposed in spirit only (there is no bounding-box regression target in a classification setting): each fused level P_i is adaptive-average-pooled to a fixed 2×2 grid (`ANCHOR_GRID=2`) rather than to a single global-average-pooled scalar, giving four fixed spatial “anchor” descriptors per level that are concatenated.
- **Per-level quantum registers:** each pyramid level’s pooled descriptor receives its own PCA→shrinkage-LDA reduction and its own `QUBITS_PER_LEVEL`-qubit quantum register (rather than being flattened into one shared PCA/LDA vector), so the joint state is $|\psi_{\text{fine}}\rangle \otimes |\psi_{\text{mid}}\rangle \otimes |\psi_{\text{global}}\rangle$ on disjoint wire subsystems.
- **Per-level conv/pool:** each register is compressed by its own small QCNN conv/pool stack (reusing the Sec. 3.3.1 blocks unchanged) down to one surviving qubit per level.
- **Cross-level entangling fusion bridge:** a ring of CRZ + CRX gates between consecutive surviving qubits acts as the quantum stand-in for FPN’s additive top-down lateral fusion. The report is explicit that this is *not* the same operation as classical addition: two independently-encoded qubits begin in a product state with no shared information, and only the (learned) entangling gates correlate them — there is no guaranteed-safe identity/no-op the way +0 is safe classically.
- **Shared classifier head:** the same `StronglyEntanglingLayers` readout used in the single-scale model reads out the fused, multi-scale state.

3.6.3 Configuration

`QUBITS_PER_LEVEL=2`, `N_LEVELS=3` (fine/mid/global), for 6 total register qubits (against an 8-qubit single-scale budget); encoding fixed to `Pi4Angle`; classical skip connection reuses only the global-level anchor vector (8 dimensions); 30 epochs with the same reduce-LR-on-plateau schedule as the single-scale model.

3.6.4 Result

The classical FPN feature extractor backbone reached 62.7% own-head validation accuracy. The resulting FPN-QCNN model reached only **54.3%** test accuracy for `Pi4Angle` — *below* even the weakest single-scale encoding in Table 3.2 (IQP, 58.0%), and well below the single-scale `Pi4Angle`-equivalent ceiling. This under-performance, and its diagnosed causes, motivate the follow-up experiment (Chapter 4).

Chapter 4

Experiment II — Refined Hybrid CNN + Residual QCNN (v6.1): Diagnosing and Repairing the FPN-QCNN Under-Performance

4.1 Overview

This experiment (notebook `QCNN_CIFAR10_Hybrid_v6_1.ipynb`) re-runs the Experiment I pipeline with two changes: (a) the classical CNN feature extractor’s training budget is raised from 25 to 40 epochs, and (b) the FPN-QCNN multi-scale variant is reconfigured based on an explicit root-cause diagnosis of why it under-performed the single-scale baseline in Experiment I. All architectural components (encodings, QCNN conv/pool blocks, classifier head) are otherwise identical to Chapter 3.

4.2 Single-scale re-benchmark with a longer CNN training budget

Raising `CNN_EPOCHS` from 25 to 40 allowed the CNN feature extractor to reach a higher own-head best validation accuracy of 67.8% (vs. 63.7% in Experiment I), and PCA(16) on the resulting embedding explained 94.9% of variance (vs. 96.0% previously). Despite the stronger backbone, downstream QCNN test accuracy changed only marginally (Table 4.1), suggesting the pipeline’s accuracy ceiling at this sample size is governed more by the PCA→LDA→QCNN stages than by CNN backbone quality alone.

Table 4.1: Experiment II single-scale hybrid CNN+QCNN test accuracy, longer CNN training budget (compare with Table 3.2).

Encoding	v6 test acc.	v6.1 test acc.	Δ
AQHE-DualEnc	0.620	0.615	−0.005
FRQI	0.594	0.607	+0.013
IQP	0.580	0.590	+0.010

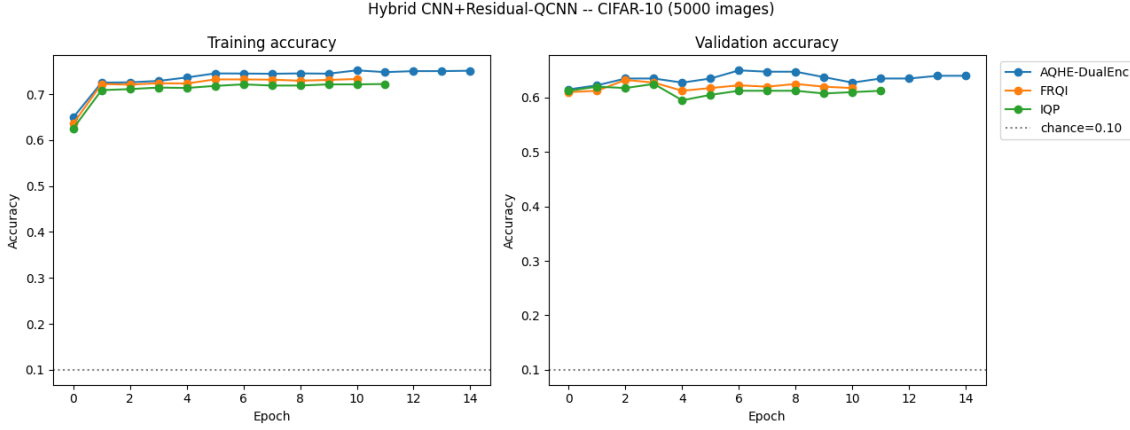


Figure 4.1: Training and validation accuracy curves, Experiment II single-scale re-benchmark (longer CNN training budget). Dotted grey line marks chance level (0.10).

4.3 Root-cause diagnosis of the FPN-QCNN under-performance

Four causes were identified for why the Experiment I FPN-QCNN (54.3%) under-performed the single-scale baseline (best 62.0%, weakest benchmarked encoding still 58.0%):

1. **Information starvation.** `preprocess_level`'s per-level LDA step is capped at `out_dim = QUBITS_PER_LEVEL`. With `QUBITS_PER_LEVEL=2`, each of the three pyramid levels could retain only 2 discriminative dimensions, so only $3 \times 2 = 6$ total dimensions reached the circuit — *fewer* than the single-scale baseline's $\text{PCA}(16) \rightarrow \text{LDA}(8) = 8$ total dimensions, despite the “multi-scale” framing suggesting richer input.
2. **Single-scale classical skip.** The classical residual/skip connection reused only the global-level anchor vector a_3 , so the “multi-scale” claim did not even hold for the classical half of the final feature vector.
3. **Shallow fusion bridge.** A single CRZ+CRX pass across the three surviving qubits is one layer of pairwise interaction, versus the single-scale baseline's full `StronglyEntanglingLayers` operating jointly over 4 register qubits *before* any information is discarded by per-branch pooling. In FPN-QCNN, each level is pooled down to 1 qubit *before* any cross-level interaction happens at all, so within-level correlations are compressed away before fusion even starts.
4. **Aggressive learning-rate decay locking in a plateau.** Both models plateaued in *training* accuracy (not validation accuracy) with no sign of overfitting (val tracked train closely) — i.e. genuine underfitting, not memorization. The FPN-QCNN's learning rate was halved every time two epochs passed without a new best, reaching $16\times$ smaller than its starting value by epoch 20, while training accuracy had been flat since roughly epoch 11. This aggressive, early shrinkage likely removed the step size needed to escape the plateau.

4.4 Fixes applied and re-run configuration

- `QUBITS_PER_LEVEL`: $2 \rightarrow 3$, giving $3 \times 3 = 9$ total register dimensions — now *exceeding* the single-scale baseline's 8.
- Classical skip connection widened to pool a slice from all three anchor vectors (a_1, a_2, a_3) at equal total width, rather than only the global-level vector.

- **FPN_BRIDGE_ROUNDS:** 1 $\rightarrow$ 2 full entangling rounds across the surviving qubits before the classifier head.
- **Explore-then-shrink learning-rate schedule:** on a non-improving epoch, the learning rate is now *increased* (factor 1.4, capped at 0.1) for up to 6 consecutive non-improving epochs since the last best value, before falling back to the previous halving-based decay if that exploration budget is exhausted without a new best. Epoch budget lowered from 30 to 20 per the notebook’s stated scope constraints.

4.5 Result: v6.1 FPN-QCNN

Table 4.2: FPN-QCNN test accuracy before and after the Sec. 4.3 fixes.

Model	Total register qubits	Total encoded dims	Test acc.
Single-scale baseline (v6.1, AQHE-DualEnc)	8	8	0.615
FPN-QCNN v1 (Experiment I)	6	6	0.543
FPN-QCNN v1.1 (post-fix, Experiment II)	9	9	0.465

Contrary to the diagnosis’s intent, the corrected FPN-QCNN configuration performed *worse* (46.5%) than both the original FPN-QCNN attempt (54.3%) and the single-scale baseline (61.5%–62.0%). Inspection of the training log shows the explore-then-shrink schedule pushed the learning rate up to its cap (0.1) without finding a new best validation accuracy, and training accuracy itself declined during the explore phase (from 0.521 at epoch 4 to 0.494–0.517 over epochs 5–10) before early stopping recovered the best checkpoint (validation accuracy 0.487, epoch 3). This indicates the increased learning rate was pushed too high for the FPN-QCNN loss landscape, causing oscillation rather than escape from a genuine plateau. As the notebook’s own diagnosis anticipates, this negative result is itself informative: it suggests the single-scale ceiling observed across v3–v5 was not purely an information-bottleneck effect, and that deepening the *single-scale* circuit (more qubits, more entangling layers) is a more promising direction than continued multi-scale pyramid fusion at the qubit budgets tractable here.

Chapter 5

Experiment III — Learnable Quanvolutional Feature Extraction

5.1 Overview

This experiment (notebook `QCNN_CIFAR10_LearnableQuanvolution_v1.ipynb`) replaces the classical CNN feature extractor used in Experiments I–II with a fully *trainable*, patch-based quantum convolutional (“quanvolutional”) layer that is optimized end-to-end, jointly with the downstream QCNN classifier, via a single Adam optimizer over every parameter group. The explicit, stated purpose is not to claim a quantum feature extractor beats a trained classical CNN on natural images, but to measure how much accuracy a fully quantum-native local-to-global feature path can recover relative to the classical-CNN pipeline, under a matched ablation-style methodology.

5.2 Dataset and preprocessing

- **Dataset:** CIFAR-10, 150 images/class (1,500 total) — smaller than Experiments I–II’s 500/class because this pipeline performs many more quantum circuit evaluations per image per forward pass.
- **Split:** 1,080 train / 120 val / 300 test.
- **Classical (fixed, non-trainable) patch pre-filtering:** each 32×32 image is tiled into non-overlapping 8×8 patches ($4 \times 4 = 16$ patches/image). Within each patch, three channels are computed and each reduced to a 2×2 (= 4-value) sub-grid:
 - *intensity* — mean grayscale luminance ($0.299R + 0.587G + 0.114B$);
 - *edge* — mean finite-difference gradient magnitude;
 - *texture* — local variance of luminance (scaled $\times 4$ before clipping, since variance occupies a much smaller numeric range than the mean-based channels).

All channel values are scaled to $[0, \pi]$ for angle encoding.

5.3 Architecture

5.3.1 Learnable quanvolution layer

For each of the three channels (intensity/edge/texture), a small trainable quantum circuit (built from the same `build_qcnn_circuit` primitives as Experiments I–II: 4-qubit `Pi4Angle` encoding, 1 convolution layer, pooling down to 2 surviving qubits, Z-only readout) is applied to

every one of the 16 patches, with weights *shared across all patch positions within a channel* (the quantum analogue of a classical convolutional filter’s weight-sharing across spatial locations). An optional second-layer “fusion” group can mix the three channels together per pooled cell (`QUANV_NUM_LAYERS=2`); this experiment uses `QUANV_NUM_LAYERS=1` (no fusion stage) as its default, tractable configuration.

5.3.2 Downstream classifier

The pooled, concatenated quanvolutional features (24-dimensional: 16 patches $\times$ pooling $\rightarrow$ 3 channels $\times$ per-channel embedding) feed a downstream QCNN classifier, structurally identical to Experiments I–II’s classifier head (4-qubit chunks, `Pi4Angle` encoding, 1 FC layer, multi-basis readout, residual reupload scale 0.3), split into 6 classifier chunks.

5.3.3 Joint end-to-end optimization

All parameters — the per-channel quanvolution weights, the (unused, in this run) fusion weights, and the classifier weights — are optimized *jointly* in a single `qml.AdamOptimizer` loop, using the same reduce-LR-on-plateau / L2 / early-stopping recipe as Experiments I–II (learning rate 0.02, decay factor 0.5, patience 2, L2 $\lambda = 0.006/\lambda_{\text{circuit}} = 0.0005$, early-stop patience 6, batch size 32, up to 15 epochs).

5.4 Computational cost

Because each quanvolutional forward pass evaluates 54 quantum circuits per image (16 patches $\times$ 3 channels + 6 classifier chunks), the notebook’s own runtime estimator projected a full run at this configuration would require on the order of 1,630 simulated GPU/CPU-hours — this estimate is explicitly flagged in the notebook as order-of-magnitude only, anchored to per-call throughput measured in Experiment I. The actual observed training time for the 1,500-image, 15-epoch run reported below was approximately 28,556 seconds (≈ 7.9 hours).

5.5 Results

Table 5.1: Experiment III learnable quanvolution + QCNN classifier results (1,500 images, chance = 10%).

Metric	Value
Best validation accuracy (checkpoint restored)	0.267
Held-out test accuracy	0.260
Classical-CNN pipeline reference (Experiment I/II, different sample count 500/class)	0.635
Chance level	0.100

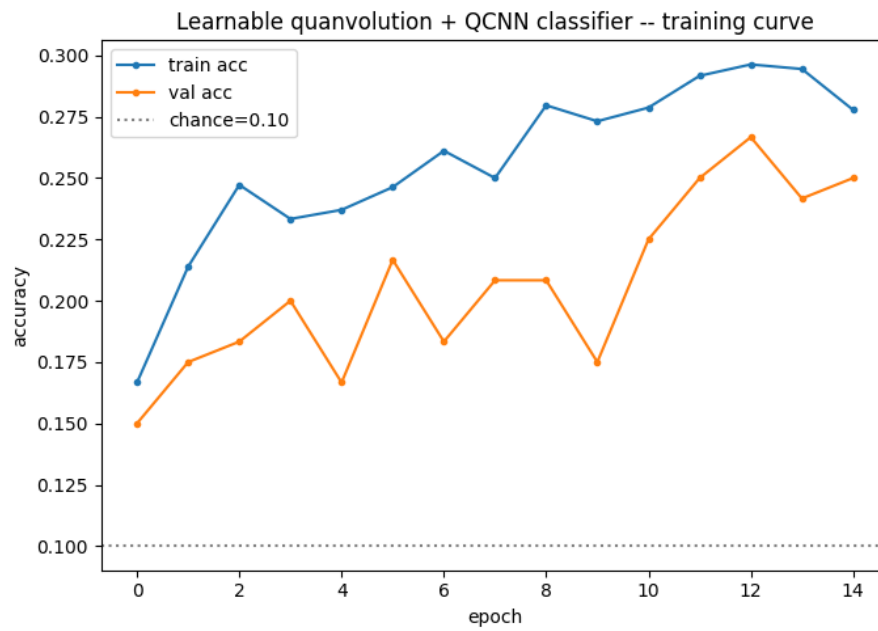


Figure 5.1: Training and validation accuracy curve, Experiment III learnable quanvolution + QCNN classifier. Dotted grey line marks chance level (0.10).

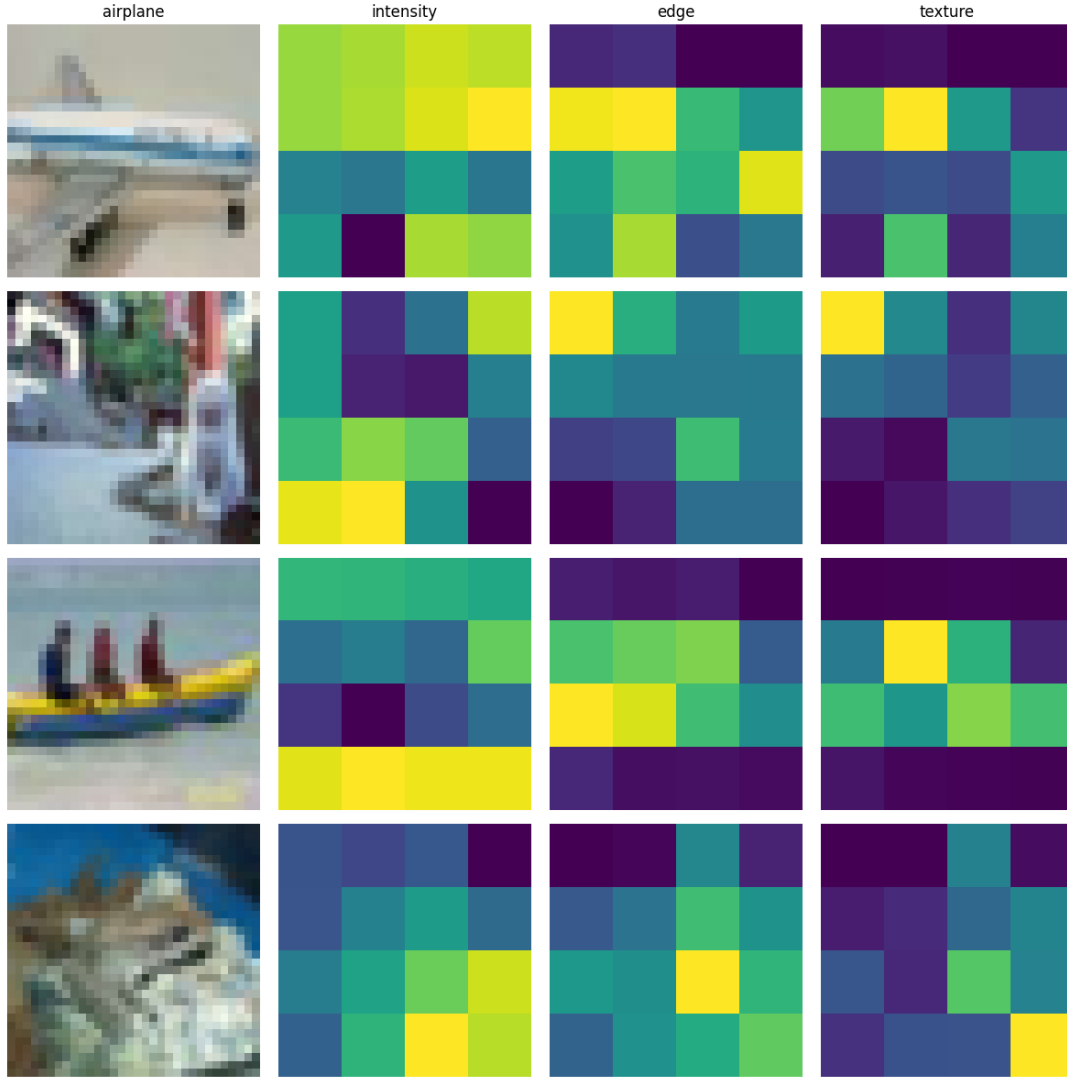


Figure 5.2: Qualitative check: raw (pre-quantum, classical) intensity/edge/texture patch grids for four representative test images, used to sanity-check what information the trainable quantum channels are actually operating on.

The notebook explicitly cautions that this 26.0%–26.7% result is *not* a like-for-like comparison against the 63.5% reference (different sample counts, epochs, and architectures), and recommends re-running the classical-CNN pipeline at the same `SAMPLES_PER_CLASS` for a fair comparison. Under the reported configuration, the learnable quanvolutional path recovers well above chance but falls substantially short of the classical-CNN-based hybrid pipeline. Contributing factors identified in the notebook’s own summary and diagnosis include: the small sample size (150/class) relative to the model’s patch-level parameter count; the shallow single-layer quanvolution (no fusion stage mixing channels); and the same family of information-bottleneck effects diagnosed for FPN-QCNN in Sec. 4.3 (compressing $16 \text{ patches} \times 3 \text{ channels}$ down to a 24-dimensional embedding before classification is itself a significant bottleneck one level earlier in the pipeline).

5.6 Suggested next steps (as recorded in the notebook)

- Re-run the classical-CNN pipeline at the identical `SAMPLES_PER_CLASS=150`, `EPOCHS=15` budget for a genuinely fair comparison;

- If enabling the two-layer fusion stage underperforms the single-layer variant, inspect the fusion-stage embedding dimensionality for the same information-starvation failure mode diagnosed for FPN-QCNN;
- Scale `SAMPLES_PER_CLASS` toward 500 and/or enable multi-basis (`QUANV_MULTI_BASIS=True`) readout for a richer (but $3\times$ costlier) per-patch feature space once runtime allows;
- Freeze the quanvolution weights at initialization (skip them from the optimizer) as a controlled ablation, to isolate how much of the achieved accuracy is attributable to *learning* the quantum filters versus using them as fixed random projections [3].

Chapter 6

Experiment IV — Learnable Quanvolution with Raw RGB Channels and a Second Fusion Layer (Two-Level Quantum Feature Pyramid)

6.1 Overview

This experiment (notebook `QCNN_CIFAR10_LearnableQuanvolution_v1_-_Copy.ipynb`) is a direct follow-up to Experiment III (Chapter 5). It keeps the same overall recipe — a fully learnable, patch-based quanvolutional feature extractor trained jointly with a downstream QCNN classifier via a single Adam optimizer — but changes two things simultaneously: (i) the three classically-derived input channels are switched from intensity/edge/texture to *raw R, G, B colour channels*, and (ii) a second, trainable **fusion quanvolution layer** is enabled (`QUANV_NUM_LAYERS=2`), which mixes the three per-channel embeddings together at every pooled spatial cell before the result is handed to the classifier. Conceptually this is a two-level quantum feature pyramid: level 1 extracts per-colour-channel local features patch-by-patch (weight-shared across the 16 patches, exactly as in Experiment III), and level 2 fuses those per-channel features across colour at each pooled location, mirroring how a second classical convolutional layer would combine the output feature maps of a first layer. As in Experiment III, every quantum circuit — per-channel, fusion, and classifier — is built from the same `build_qcnn_circuit` primitives reused throughout this report, so no new, unvalidated circuit code is introduced; only the input channels and the number of quanvolution layers change relative to Experiment III.

6.2 Dataset and preprocessing

- **Dataset:** CIFAR-10, 150 images/class (1,500 total, identical sampling budget to Experiment III), split into 1,080 train / 120 val / 300 test images (stratified, seed 42).
- **Classical (fixed, non-trainable) patch extraction:** each 32×32 image is tiled into $4 \times 4 = 16$ non-overlapping 8×8 patches. Within each patch, the raw R, G, and B channels (normalized to $[0, 1]$) are each reduced to a 2×2 (= 4-value) sub-grid via 2×2 sub-block averaging, then scaled to $[0, \pi]$ for angle encoding — i.e. the patch grid, sub-block reduction, and $[0, \pi]$ rescaling are unchanged from Experiment III; only the three channels’ *content* changes, from intensity/edge/texture to raw R/G/B.
- **Assumption/inconsistency worth flagging:** the notebook’s own Section 4 markdown cell still describes the three channels as “intensity”, “edge”, and “texture” (carried over,

apparently unedited, from the Experiment III template) and motivates the channel choice in those terms. The executed code in this notebook, however, computes plain per-channel R, G, B means and does *not* compute any grayscale-luminance, gradient, or variance channel. The results below reflect the code that was actually run (raw RGB), not the stale markdown description; this documentation/code mismatch is reproduced here for transparency rather than silently corrected.

6.3 Architecture

6.3.1 Level-1 quanvolution: per-channel, patch-shared circuits

For each of the three channels (R, G, B), a small trainable quantum circuit (4-qubit `Pi4Angle` encoding, 1 convolution layer, pooling down to `QUANV_N_OUT_PATCH=2` surviving qubits, Z-only readout) is applied to all 16 patches, with weights shared across patch positions within a channel — identical in structure to Experiment III’s per-channel circuits.

6.3.2 Level-2 quanvolution: the fusion layer

The pooled per-patch grid (4×4) is itself pooled by `QUANV_POOL_SIZE=2` down to a $2 \times 2 = 4$ -cell grid. At each of these four pooled cells, a second trainable quantum circuit (the “fusion” group) takes the three channels’ per-cell embeddings as input and produces a `QUANV_FUSION_N_OUT=2`-qubit readout, i.e. one fusion circuit evaluation per pooled cell, mixing colour-channel information that was previously kept in separate, non-interacting circuits. This is the component that was present but disabled (`QUANV_NUM_LAYERS=1`) in Experiment III.

6.3.3 Downstream classifier

The fused, concatenated features (embedding dimension 26) are split into 7 classifier chunks of 4 qubits each and fed to a downstream QCNN classifier structurally identical to every other experiment’s classifier head in this report (`Pi4Angle` encoding, 1 FC layer, multi-basis readout, residual reupload scale 0.3).

6.3.4 Joint end-to-end optimization

All five parameter groups — R, G, B, fusion, and classifier, in that flat-parameter order — are optimized jointly in a single `qml.AdamOptimizer` loop, with the same reduce-LR-on-plateau / L2 / early-stopping recipe as Experiment III (learning rate 0.02, decay factor 0.5, decay patience 2, L2 $\lambda = 0.006$ / $\lambda_{\text{circuit}} = 0.0005$, early-stop patience 6, batch size 32, budgeted for up to 15 epochs).

6.4 Computational cost

Adding the fusion layer increases the per-sample circuit-call count from Experiment III’s 54 to 59 (16 patches $\times$ 3 channels = 48 per-channel calls, +4 fusion calls, one per pooled cell, +7 classifier-chunk calls). The notebook’s own runtime estimator projected an order-of-magnitude full-run cost of roughly 1,781 simulated hours at this configuration — again explicitly flagged as a rough anchor, not a precise forecast. The actual observed training run took 21,317.9 seconds (≈ 5.9 hours) and terminated via **early stopping after epoch 10 of the budgeted 15**, once validation accuracy had not improved for 6 epochs despite three learning-rate decays ($0.02 \rightarrow 0.01 \rightarrow 0.005 \rightarrow 0.0025$).

6.5 Results

Table 6.1: Experiment IV (RGB channels, two-level fusion quanvolution) results, 1,500 images, chance = 10%, compared against Experiment III (intensity/edge/texture, single layer).

Metric	Exp. IV (RGB + fusion)	Exp. III (single layer)
Best validation accuracy (checkpoint restored)	0.250	0.267
Held-out test accuracy	0.217	0.260
Classical-CNN pipeline reference (different sample count 500/class)	0.635	0.635
Chance level	0.100	0.100
Circuit calls per sample	59	54
Observed training wall-clock	≈ 5.9 h (early-stopped, epoch 10/15)	≈ 7.9 h (full 15 epochs)

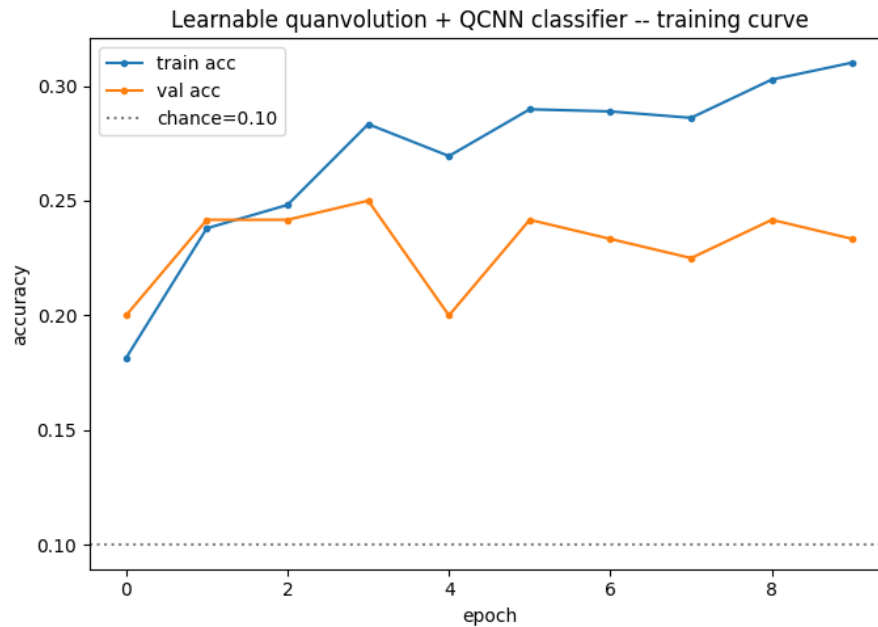


Figure 6.1: Training and validation accuracy curve, Experiment IV (RGB channels, two-level fusion quanvolution + QCNN classifier). Dotted grey line marks chance level (0.10). Training halts at epoch 10 due to early stopping.

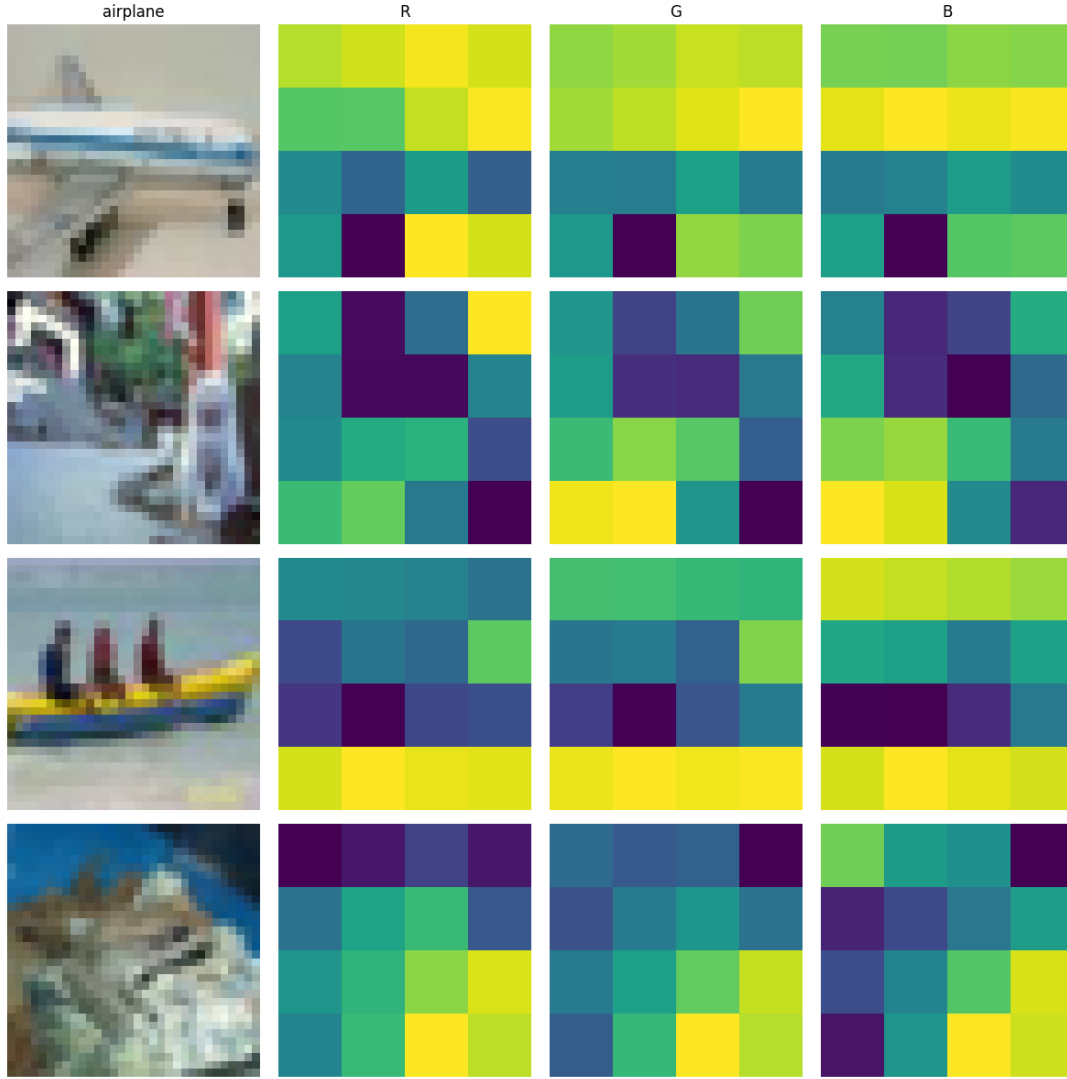


Figure 6.2: Qualitative check: raw (pre-quantum, classical) R, G, B patch grids for four representative test images, used to sanity-check what information the trainable quantum channels are actually operating on in this RGB variant.

Contrary to the intuition that (a) raw colour information and (b) an extra fusion stage should each give the classifier strictly more to work with, this configuration under-performs Experiment III on both validation (25.0% vs. 26.7%) and test accuracy (21.7% vs. 26.0%), while costing more compute per sample and still not completing its epoch budget before early stopping triggered. Both variants remain well above the 10% chance level and well below the 63.5% classical-CNN reference (again explicitly not a like-for-like comparison at this sample size). Plausible contributing factors, read against the diagnosis already established for FPN-QCNN (Sec. 4.3) and for Experiment III itself:

- **An additional bottleneck stage, not additional capacity.** The fusion layer compresses three independent 2-qubit channel readouts down to a 2-qubit fusion readout *per pooled cell*, discarding roughly a third of the per-channel information at every one of the four pooled cells before the classifier ever sees it — the same information-starvation pattern diagnosed for the FPN-QCNN pyramid and for Experiment III’s single-layer bottleneck, now stacked a second time.
- **Raw RGB may be a harder input than intensity/edge/texture for a 4-qubit,**

single-Pi4Angle-layer circuit. Intensity, edge, and texture channels are explicitly engineered to bias the circuit toward brightness, boundary, and high-frequency structure respectively; raw R, G, B channels are highly redundant with one another for natural images (they are strongly correlated per-pixel), so three separate 4-qubit circuits processing near-duplicate information may add optimization difficulty without adding much genuinely new signal.

- **More trainable parameters at an unchanged, small sample size.** The fusion layer adds a fourth quantum-circuit parameter group without any increase to the 1,500-image dataset, worsening the parameters-to-samples ratio that Experiment III's own summary already flagged as a limiting factor.
- **Early stopping cut training short.** Training halted at epoch 10 of 15 once the plateau-and-decay criterion was met, so it remains possible — though not established by this run — that a longer patience budget, a lower initial learning rate, or a warm-start from Experiment III's already-trained per-channel weights would let the fusion stage recover some of the gap.

6.6 Suggested next steps (as recorded in the notebook)

- Re-run the classical-CNN pipeline at the identical `SAMPLES_PER_CLASS=150`, `EPOCHS=15` budget for a genuinely fair comparison against both quanvolution variants;
- Since `QUANV_NUM_LAYERS=2` under-performed `QUANV_NUM_LAYERS=1` (Experiment III), inspect the fusion-stage embedding dimensionality for the same information-starvation failure mode diagnosed for FPN-QCNN;
- Scale `SAMPLES_PER_CLASS` toward 500 and/or enable multi-basis (`QUANV_MULTI_BASIS=True`) readout for a richer (but $3\times$ costlier) per-patch feature space once runtime allows;
- As a controlled ablation, freeze the quanvolution channel weights after initialization (excluding them from the optimizer's parameter list) to isolate how much of the achieved accuracy is attributable to *learning* the quantum filters versus using them as fixed random projections [3];
- Reconcile the notebook's Section 4 markdown description (intensity/edge/texture) with its actual implementation (raw RGB) before further iterations, to avoid future ambiguity about which channel definition a given result corresponds to.

Chapter 7

Experiment V — Unified Benchmark of 18 Quantum Encodings with Pauli Correlation Encoding

7.1 Overview

This experiment (notebook `QML_CIFAR10_v3_PCE_fixed.ipynb`) is the third iteration of a broader QML encoding benchmark and is architecturally independent of the CNN-based hybrid pipeline in Experiments I–III: rather than a trainable classical CNN front end, it uses `StandardScaler`→`PCA`→(`LDA`) directly on flattened raw RGB pixels, and evaluates each encoding both as a **quantum kernel** for a classical multiclass SVM and, for one encoding, as a **variational quantum classifier** (VQC). Its headline contribution is the addition of an 18th encoding, **Pauli Correlation Encoding (PCE)** [7], alongside the seventeen encodings carried over (with several bug-fixed corrections) from the notebook’s v1/v2 predecessors.

7.2 Dataset and dual preprocessing pipelines

- **Dataset:** CIFAR-10, 60 images/class (600 total), loaded via `torchvision` with fallbacks to `tensorflow.keras` and `sklearn`’s `fetch_openml` (and, failing all three, a Digits-dataset proxy for pipeline smoke-testing). Split 75%/25% train/test *before* any preprocessing, to avoid train/test leakage through the fitted PCA/LDA/scaler transforms.
- **Pipeline A (17 standard encodings):** RGB (3072-d, flattened) →`StandardScaler`→`PCA`(50, `whitened`)→`LDA`(8) (supervised)→`MinMaxScaler`[0, π]. On the 600-image subsample, PCA-50 retained 86.5% of variance.
- **Pipeline B (PCE only):** RGB →`StandardScaler`→`PCA`(64, `whitened`)→`MinMaxScaler`[0, π] (no LDA stage, since PCE’s hardware-efficient ansatz needs $m = 64$ continuous features, far beyond LDA’s $K - 1 = 9$ cap). PCA-64 retained 89.0% of variance — roughly 7× more retained information than a PCA-4 front end used in the notebook’s v1 predecessor.

Figure 7.1 reproduces the notebook’s own dual-pipeline architecture diagram.

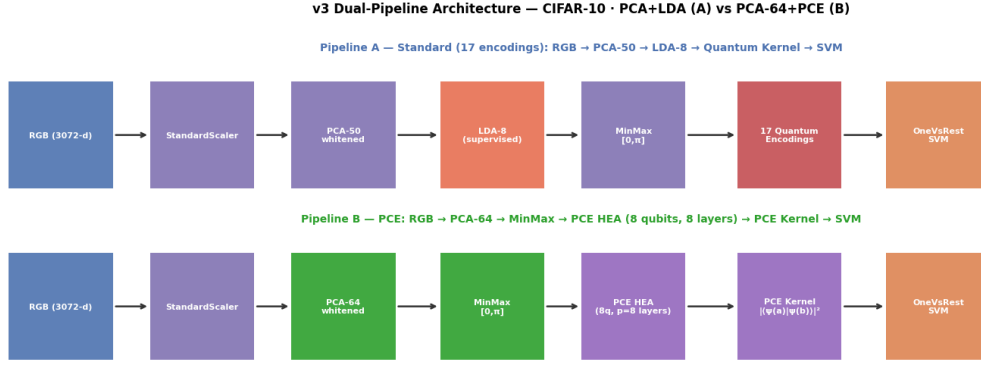


Figure 7.1: Dual-pipeline architecture, Experiment V. Pipeline A (top) feeds 17 standard encodings via PCA-50→LDA-8; Pipeline B (bottom) feeds Pauli Correlation Encoding via PCA-64 directly.

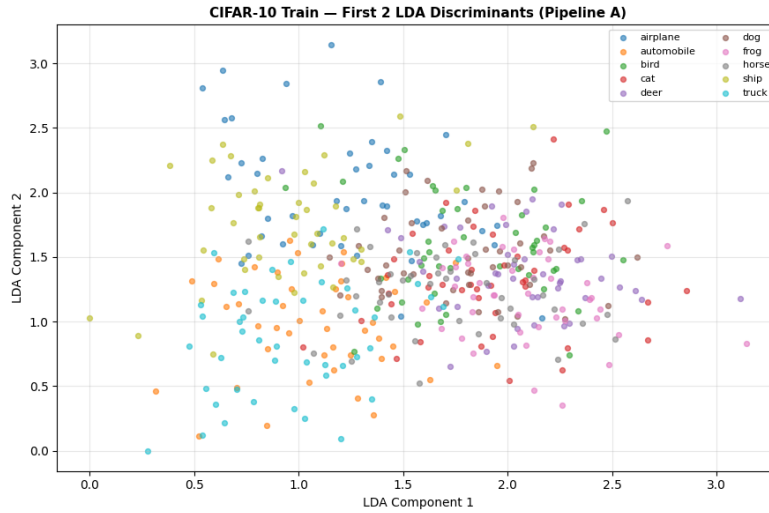


Figure 7.2: First two LDA discriminant components of the CIFAR-10 training subsample (Pipeline A), colour-coded by class.

7.3 The eighteen quantum encodings

7.3.1 Seventeen standard encodings

Table 7.1 summarizes the seventeen carried-over encodings. Several were explicitly corrected relative to the notebook’s v1/v2 predecessors: IQP was extended from nearest-neighbour-only IsingZZ interactions to all-pairs R_{ZZ} terms across two full layers [4]; QHE was corrected to the canonical Hadamard→PhaseShift→inverse-QFT construction; DenseAngle gained genuine cross-term $R_Z(x_i x_j)$ interactions; and FRQI’s controlled-rotation wire indexing was corrected to a circular pattern on a uniform superposition [8].

Table 7.1: The seventeen standard quantum encodings benchmarked in Experiment V (grouped by family).

Family / Encoding	Description
<i>Angle / basis encodings</i>	
Rotational, SimpleAngle	Direct single-axis rotation encoding of each feature.
Pi4Angle	Hadamard superposition + scaled R_Y rotation.
EntangledAngle	Angle rotations plus a linear entangling CNOT chain.
Basis	Computational-basis (bit-string) encoding.
Amplitude	Amplitude encoding of the normalized feature vector into the statevector.
DenseAngle	$R_Z(x_i^2)$ plus CNOT-sandwiched cross-term rotations.
ShotBased	Finite-shot-sampled variant of an angle encoding.
<i>Image / kernel / AQHE encodings</i>	
FRQI	Flexible Representation of Quantum Images [8].
NEQR	Novel Enhanced Quantum Representation of digital images [9].
IQP	Instantaneous Quantum Polynomial-time circuit, all-pairs R_{ZZ} , 2 layers [4].
QHE	Quantum Helix Encoding: Hadamard + PhaseShift + inverse QFT.
Hamiltonian	Trotterized evolution under an Ising-like Hamiltonian.
GraphEncoding	Graph-structured entangling encoding.
AQHE-LogHelix	Logarithmically-warped phase placement + inverse QFT.
AQHE-FibSpiral	Golden-angle spiral phase placement + inverse QFT.
AQHE-DualEnc	Combined amplitude + logarithmic-phase encoding + inverse QFT.

7.3.2 Pauli Correlation Encoding (PCE)

PCE encodes $m = 64$ continuous PCA features onto $n = 8$ qubits via a hardware-efficient ansatz (HEA) with $p = \lfloor m/n \rfloor = 8$ layers, each layer applying one $R_Y(x)$ rotation per qubit followed by a linear chain of CZ entangling gates, plus one final R_Y layer: $(p + 1) \times n = 72$ single-qubit rotation gates and $p \times (n - 1) = 56$ two-qubit gates, for 128 gates total on a $2^8 = 256$ -amplitude statevector. The resulting statevector overlap $K(x_a, x_b) = |\langle \psi(x_a) | \psi(x_b) \rangle|^2$ is used directly as a quantum kernel. The encoding's theoretical capacity is $m_{\max}(n, k) = \binom{n}{k} \times 3^k$; for $n = 8, k = 2$: $m_{\max} = \binom{8}{2} \times 9 = 252 \geq 64$, confirming the chosen configuration can in principle encode all 64 PCA features (Figure 7.3). For comparison, the notebook notes that a QAOA-style approach encoding a comparable number of variables would require on the order of 42,000+ gates, versus PCE's 128.

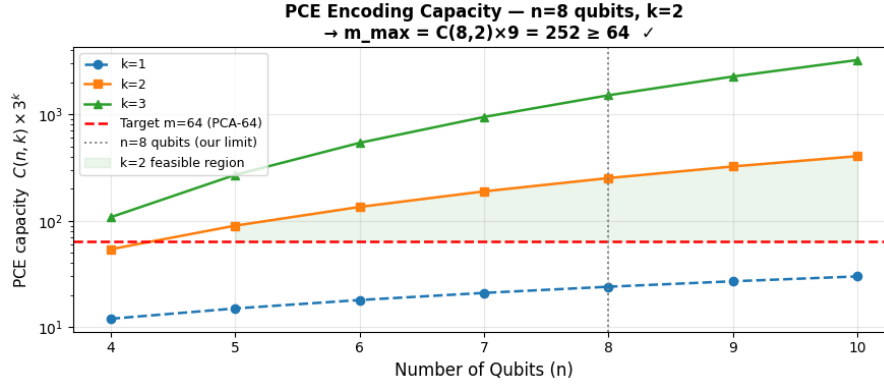


Figure 7.3: PCE encoding capacity $m_{\max}(n, k) = \binom{n}{k} \times 3^k$ as a function of qubit count n , for correlation orders $k \in \{1, 2, 3\}$. The chosen operating point ($n = 8, k = 2$) lies inside the feasible region for the target $m = 64$ features.

7.4 Classifiers: quantum-kernel SVM and VQC

- **Quantum-kernel multiclass SVM:** for each encoding, a full train×train and test×train kernel matrix is built from pairwise circuit-state overlaps, and a One-vs-Rest SVM with a precomputed kernel ($C = 1.0$) is fit on top.
- **Variational Quantum Classifier (VQC):** a 2-layer variational circuit trained by gradient descent (used selectively, since VQC training is far more expensive than one-shot kernel evaluation; the main 18-encoding benchmark uses `RUN_VQC=False` and only the kernel-SVM route, with one dedicated full VQC training run reported separately in Sec. 7.8).

7.5 Main benchmark: all 18 encodings

Table 7.2 reports One-vs-Rest quantum-kernel SVM test accuracy for all eighteen encodings on the 600-image CIFAR-10 subsample (train=450, test=150; chance = 10%).

Table 7.2: Experiment V main benchmark: quantum-kernel SVM test accuracy, all 18 encodings.

Encoding	Test acc.	Wall time (s)	Feature pipeline
AQHE-DualEnc	0.320	145.3	LDA-8
Rotational	0.300	44.9	LDA-8
SimpleAngle	0.300	45.8	LDA-8
EntangledAngle	0.300	174.2	LDA-8
FRQI	0.300	62.1	LDA-8
QHE	0.300	145.7	LDA-8
Pi4Angle	0.293	50.1	LDA-8
AQHE-LogHelix	0.280	138.5	LDA-8
DenseAngle	0.273	430.6	LDA-8
NEQR	0.267	99.4	LDA-8
Amplitude	0.260	284.5	LDA-8
AQHE-FibSpiral	0.253	146.6	LDA-8
ShotBased	0.247	71.0	LDA-8
Basis	0.207	132.1	LDA-8
PCE	0.180	274.1	PCA-64 (64-dim)
IQP	0.180	481.3	LDA-8
Hamiltonian	0.160	478.4	LDA-8
GraphEncoding	0.093	99.3	LDA-8

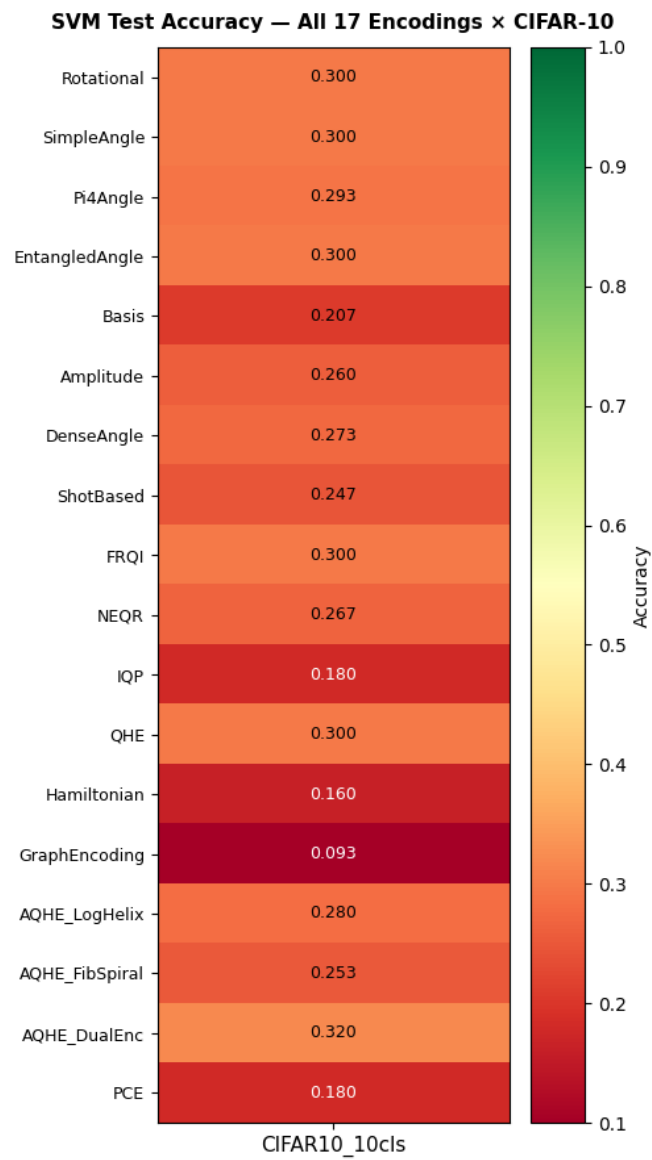


Figure 7.4: Summary heat map of quantum-kernel SVM test accuracy across all 18 encodings on CIFAR-10 (10 classes).

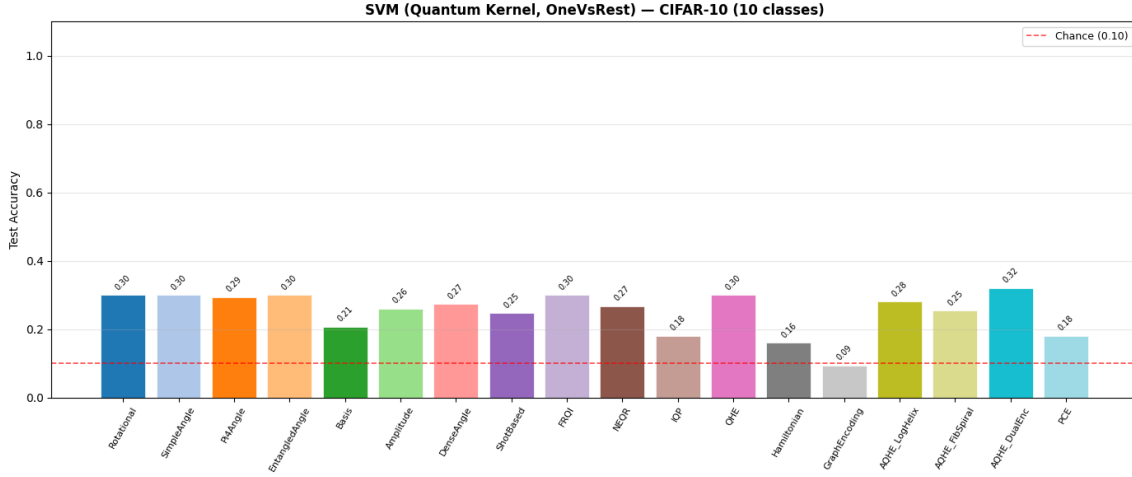


Figure 7.5: Comparative bar chart of quantum-kernel SVM test accuracy across all benchmarked encodings.

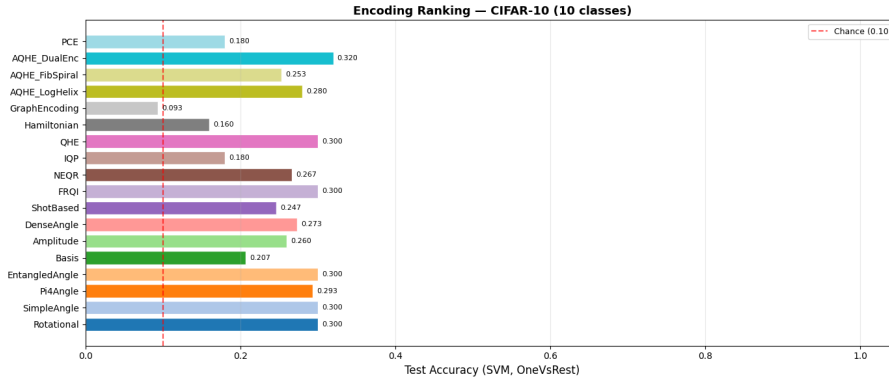


Figure 7.6: Encoding-family radar/spider chart comparing accuracy across encoding groups.

Observation. AQHE-DualEnc is again the strongest encoding (consistent with its ranking in Experiments I–II), though at this much smaller sample size (600 vs. 5,000 images) and using raw-pixel PCA/LDA rather than a trained CNN embedding, absolute accuracy is far lower (32.0% vs. 62.0%). Several encodings (Rotational, SimpleAngle, EntangledAngle, FRQI, QHE) tie closely at 30.0%. GraphEncoding and Hamiltonian perform worst, close to or below what a degenerate classifier could achieve on an imbalanced split of this size. PCE, despite seeing $8\times$ more raw input dimensions (64 vs. 8) than every other encoding, does *not* outperform the LDA-based encodings at its baseline 8-layer configuration — discussed further in Sec. 7.9.

7.6 Qubit-count sensitivity sweep

For each encoding except PCE (whose 8-qubit, 64-feature pipeline is qualitatively different from the LDA-based encodings and is not meaningfully swept this way), PCA+LDA is refit with $\text{LDA dims} = n_{\text{qubits}} \in \{2, \dots, 8\}$ and the quantum-kernel SVM is re-evaluated. Figure 7.7 shows the resulting accuracy-vs-qubit-count curves, split into an angle/basis-encoding group and an image/kernel/AQHE-encoding group.

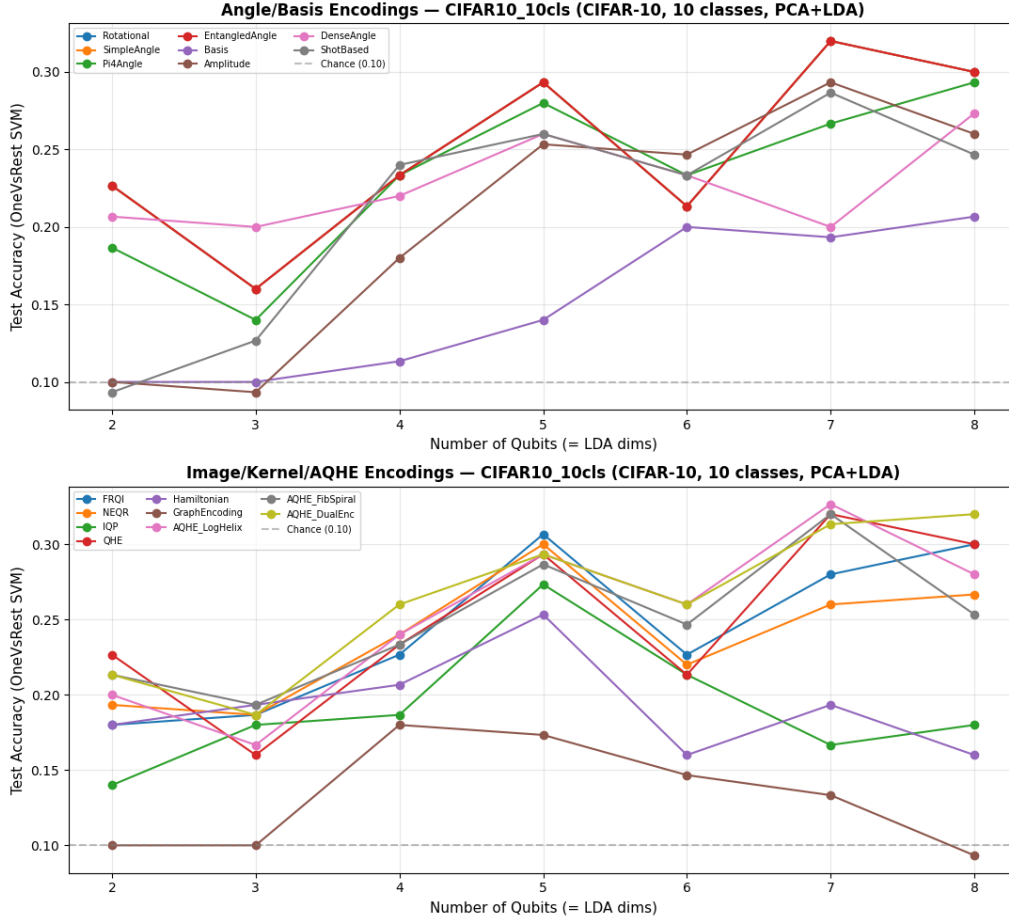


Figure 7.7: Test accuracy as a function of qubit count (= LDA dimensionality), swept from 2 to 8 qubits, for all encodings except PCE. Dashed grey line marks chance level (0.10).

At the low end (2 qubits), most encodings cluster near 0.18–0.23, with several (Basis, Amplitude, GraphEncoding) sitting at chance (0.10). Accuracy generally improves as qubit count increases toward 4, after which returns diminish and several encodings plateau or mildly regress by 8 qubits, consistent with LDA’s diminishing marginal discriminative information per additional component beyond a handful of dimensions on this sample size.

7.7 LDA dimensionality sensitivity

A focused sweep over four representative encodings (Rotational, QHE, IQP, Hamiltonian) at LDA dims $\in \{8, 16, 32, 64\}$ (capped in practice at $K - 1 = 9$) is shown in Figure 7.8.

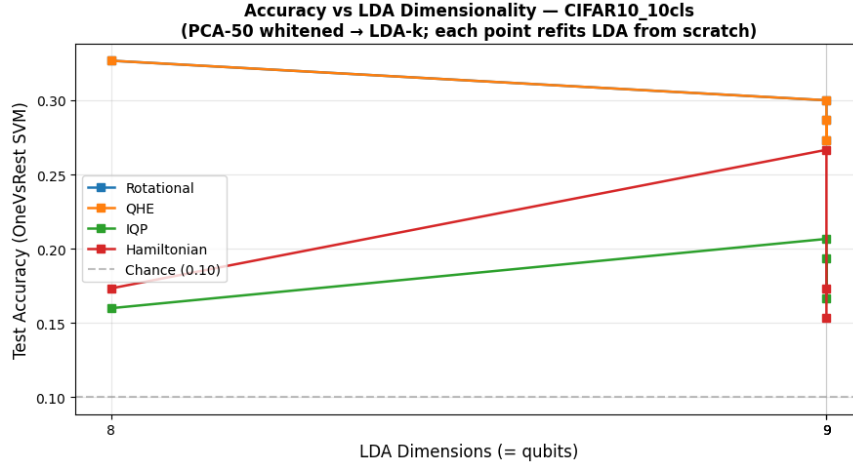


Figure 7.8: Test accuracy vs. LDA dimensionality (= qubit count) for four representative encodings; each point refits LDA from scratch. LDA is hard-capped at $K - 1 = 9$ components for $K = 10$ classes.

Rotational and QHE track each other closely across all tested dimensionalities (both peaking near 0.30–0.33 at LDA= 8), while IQP and Hamiltonian remain markedly weaker and more volatile across the sweep, consistent with their poor showing in the main benchmark (Table 7.2).

7.8 Full VQC training run

A single dedicated variational quantum classifier run (2 layers, FRQI encoding, 100 epochs, batch size 32, 360 train / 90 val / 150 test) was performed to characterize gradient-trained (as opposed to kernel-based) performance. Figure 7.9 shows the resulting training curve.

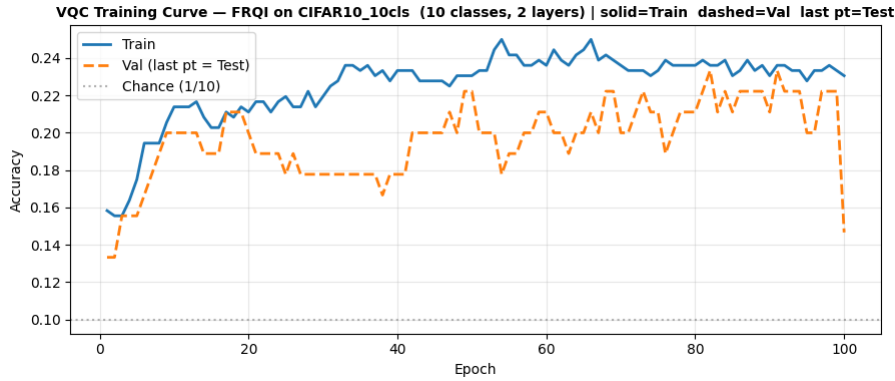


Figure 7.9: VQC (2-layer, FRQI encoding) training curve on CIFAR-10 (10 classes), 100 epochs. Solid = train, dashed = validation, final point = held-out test.

The VQC reached a final held-out test accuracy of **14.67%** — well above chance (10%) but far below FRQI’s quantum-kernel SVM accuracy on the same encoding (30.0%, Table 7.2). Both train and validation accuracy plateaued early (around epoch 30) and oscillated within a narrow band for the remaining 70 epochs, indicating the gradient-trained variational approach struggled to escape a shallow optimization landscape at this qubit count and sample size, consistent with barren-plateau-type optimization difficulties commonly reported for VQCs [6].

7.9 Refining PCE: reduced feature count and shallower ansatz (PCE v2)

Motivated by PCE’s under-performance in the main benchmark, a follow-up configuration (“PCE v2”) was tested with two changes:

1. **Supervised feature selection:** rather than using all 64 PCA dimensions, an ANOVA F-test (`sklearn.feature_selection.f_classif`, fit on train only) ranks the 64 PCA components by class-discriminative power, and only the top 24 are retained (mean F-score of retained features 3.8 vs. 2.0 across all 64).
2. **Shallower, denser ansatz:** the HEA depth is reduced from $p = 8$ layers (1 feature/qubit/layer) to $p = 2$ layers using 2 features/qubit/layer (R_Y and R_Z per qubit), followed by one linear-CZ entangling sweep per layer. This reduces analytic circuit depth from approximately 26 to approximately 6, intended to reduce kernel concentration (a known failure mode in which quantum kernel values collapse toward a constant as circuit depth grows).
3. **Hyperparameter tuning:** the SVM regularization constant C is tuned via 3-fold stratified cross-validation over $C \in \{0.5, 1.0, 2.0, 5.0, 10.0\}$ using manual kernel sub-matrix slicing (noting explicitly that `sklearn`’s built-in `cross_val_score` does not correctly slice precomputed kernels inside a `OneVsRestClassifier`).

Table 7.3: PCE ablation: baseline (v1, PCA-64, 8-layer HEA) vs. refined (v2, PCA-24 by F-score, 2-layer dual-axis HEA, tuned C), against the best LDA-based encoding.

Configuration	Test accuracy
PCE v1 (baseline, PCA-64, 8-layer HEA)	0.180
PCE v2 (PCA-24 by F-score, 2-layer HEA, tuned $C = 0.5$)	0.273
Best LDA-based encoding (AQHE-DualEnc)	0.320

The shallower, feature-selected PCE v2 configuration recovers substantially over the baseline (18.0% $\rightarrow$ 27.3%, a 9.3-point improvement) but still falls short of the best LDA-based encoding (32.0%). This is consistent with the notebook’s own stated expectation that a direct PCE-vs.-LDA-encoding comparison is not strictly apples-to-apples (PCE draws on unsupervised PCA-64 features while the other seventeen encodings draw on supervised LDA-8 features that are explicitly optimized for class separability), and suggests that PCE’s favourable gate-count and qubit-capacity scaling does not, on its own and at this problem scale, translate into a classification-accuracy advantage over a well-tuned supervised low-dimensional encoding.

7.10 Summary of observations (Experiment V)

- No single encoding dominates uniformly; AQHE-DualEnc is the strongest performer both in this experiment and in Experiments I–II, suggesting its combined amplitude+phase construction generalizes well across very different preprocessing pipelines and sample sizes.
- PCE’s large theoretical capacity ($m_{\max} = 252 \geq 64$) and low gate count (128 vs. an estimated 42,000+ for a QAOA-scale comparison) do not by themselves guarantee strong classification accuracy; circuit depth and kernel concentration, and the loss of LDA-style supervised feature selection, appear to matter more at this problem scale.

- Gradient-trained VQCs underperformed quantum-kernel SVMs using the same encoding (FRQI: 14.67% VQC vs. 30.0% kernel-SVM), consistent with known VQC trainability challenges at small sample sizes.
- Both the qubit-count sweep (Sec. 7.6) and the LDA-dimensionality sweep (Sec. 7.7) show diminishing or non-monotonic returns beyond roughly 4–8 qubits/LDA-dimensions at this 600-image sample size.

Chapter 8

Experiment VI — Hybrid CNN + Residual QCNN (v7): Full 10-Encoding Benchmark, Qubit-Count Sweep, and Quantum-vs-Classical Ablation

8.1 Overview

This experiment (notebook `QCNN_CIFAR10_Hybrid_v7.ipynb`) is the seventh iteration of the BTP’s main single-scale hybrid CNN+QCNN pipeline (Chapters 3–4) and is explicitly framed by its own changelog as “completing the encoding benchmark”: where Experiments I and II only trained the top three of ten candidate encodings to completion (owing to per-encoding wall-clock cost), v7 implements five follow-ups so that the benchmark, the qubit-count sweep, and a quantum-vs-classical ablation are all run to completion for the first time. The overall architecture — CNN feature extractor $\rightarrow$ PCA $\rightarrow$ shrinkage-LDA $\rightarrow$ residual QCNN classifier, with the same building blocks defined in Sec. 3.3.1 — is unchanged from Experiments I–II; this chapter therefore focuses on what is new: the completed benchmark, two new sweeps, and a widened FPN-QCNN encoding comparison.

8.1.1 v7 changelog

As stated in the notebook, five follow-ups were implemented after the Experiment II (v6.1) partial run, which had only benchmarked 3 of 10 encodings:

1. **Full 10-encoding sweep** (Sec. 8.3) — `N_ENCODINGS_TO_RUN` raised from 3 to 10, so every encoding defined in the notebook is now trained and tested, not just the top-3 priority subset used in Experiments I–II.
2. **Qubit-count sweep** (Sec. 8.4) — exercises `CONFIG["QUBIT_SWEEP"] = [2,3,4,5,6,7,8]`, a configuration option that existed in Experiment II’s code but was never looped over.
3. **Quantum-vs-classical ablation** (Sec. 8.5) — re-trains the best encoding with the residual re-upload and classical skip connection each independently disabled, and both disabled together, to isolate how much of the full model’s accuracy comes from the quantum circuit itself versus the classical scaffolding around it.
4. **Widened FPN-QCNN encoding sweep** (Sec. 8.6) — the multi-scale FPN-QCNN variant from Experiment II is now run for `AQHE-DualEnc` and `FRQI` in addition to `Pi4Angle`, rather than being judged on a single encoding.

5. **QCNN-vs-kernel-SVM rank correlation** (Sec. 8.7) — intended to merge this notebook’s per-encoding QCNN accuracies with the Experiment V kernel-SVM benchmark’s per-encoding accuracies and compute a Spearman rank correlation between the two architectures’ encoding preferences.

8.2 Dataset and preprocessing

Identical protocol to Experiments I–II: CIFAR-10, 500 images/class (5,000 total), split 3,600 train / 400 val / 1,000 test. The classical CNN feature extractor (same residual-block architecture as Experiments I–II, Sec. 3.3.1’s classical counterpart) was trained for up to 40 epochs with augmentation, weight decay 5×10^{-4} , and dropout 0.3, reaching a best validation accuracy of **70.0%** (higher than both Experiment I’s 63.7% and Experiment II’s 67.8% runs of the same architecture). **StandardScaler**→**PCA(16)** on the resulting 128-d embedding retained 95.0% of variance, and shrinkage LDA reduced this to the same 8 discriminant dimensions used throughout Experiments I–II. One configuration change relative to Experiment II: **FC_LAYERS** is set to 2 (one fewer **StronglyEntanglingLayers** layer in the quantum classifier head than the un-reduced value used in the very first v3 run), matching Experiment I/II’s own v3→v4 fix for over-capacity relative to the training-set size.

8.3 Full 10-encoding benchmark

All ten encodings listed in Table 3.1 were trained and tested to completion (30-epoch budget, reduce-LR-on-plateau, early stopping — identical training recipe to Experiments I–II). Table 8.1 reports held-out test accuracy.

Table 8.1: Experiment VI full 10-encoding hybrid CNN+QCNN benchmark, test accuracy (5,000 images, chance = 10%).

Encoding	Test accuracy
Pi4Angle	0.519
DenseAngle	0.511
AQHE-DualEnc	0.509
FRQI	0.507
EntangledAngle	0.505
QHE	0.503
AQHE-LogHelix	0.502
IQP	0.500
Hamiltonian	0.497
AQHE-FibSpiral	0.495

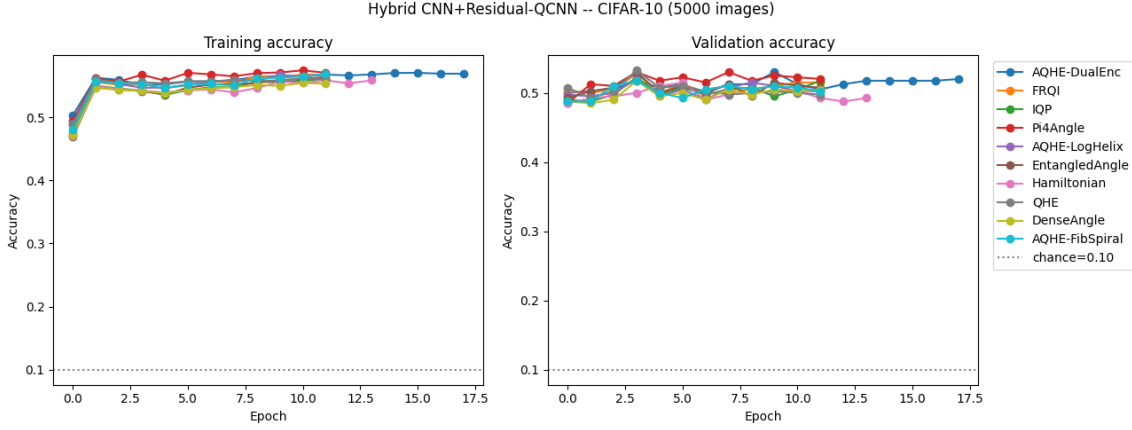


Figure 8.1: Training and validation accuracy curves for all ten benchmarked encodings, Experiment VI full sweep. Dotted grey line marks chance level (0.10).

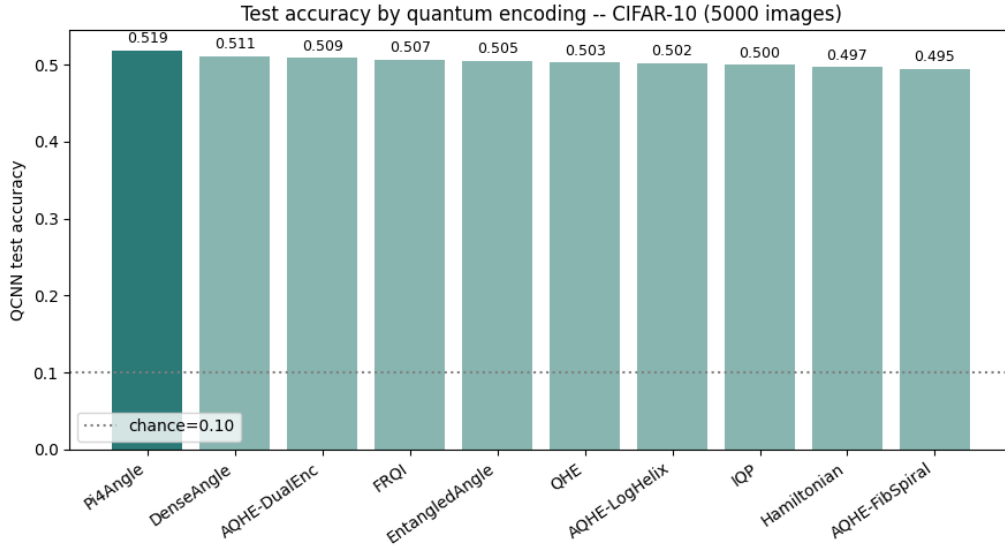


Figure 8.2: Test accuracy by quantum encoding, Experiment VI full 10-encoding sweep. Bar colour highlights the best-performing encoding (Pi4Angle).

An important discrepancy versus Experiments I–II. All ten encodings land in a narrow 0.495–0.519 band, with **Pi4Angle** — the cheapest and structurally simplest encoding benchmarked in this report — coming out on top, and **AQHE-DualEnc**, the best-performing encoding in every other experiment in this report (Experiments I, II, and V), placing only third here at 0.509. This is a materially different ranking and a materially different accuracy level from Experiments I and II, where AQHE-DualEnc reached 0.620/0.615 and even the weakest of the three encodings tested there (IQP, 0.580/0.590) clears every result in Table 8.1. No architectural change explains a gap of this size: the CNN backbone, PCA/LDA stage, QCNN building blocks, and training recipe are otherwise identical to Experiment II, and the CNN extractor in this run reached a *higher* own-head validation accuracy (70.0%) than either prior run. The most likely explanation, discussed further in Sec. 8.8, is run-to-run variance compounded with evidence of non-sequential notebook execution; this report records both sets of numbers rather than treating either as superseding the other, since neither notebook’s execution log provides enough information to determine which (if either) reflects the pipeline’s true achievable accuracy at this sample size.

8.4 Qubit-count sweep

Using the winner of the Sec. 8.3 sweep (Pi4Angle), the same 8-dimensional PCA/LDA feature vector was re-chunked into registers of $\{2, 3, 4, 5, 6, 7, 8\}$ qubits (smaller registers mean more, narrower chunks processed by the same shared circuit; larger registers mean fewer, wider chunks) and retrained end-to-end at each width.

Table 8.2: Experiment VI qubit-count sweep, Pi4Angle encoding, test accuracy vs. register size.

Qubits	2	3	4	5	6	7	8
Test acc.	0.655	0.649	0.645	0.639	0.625	0.628	0.635

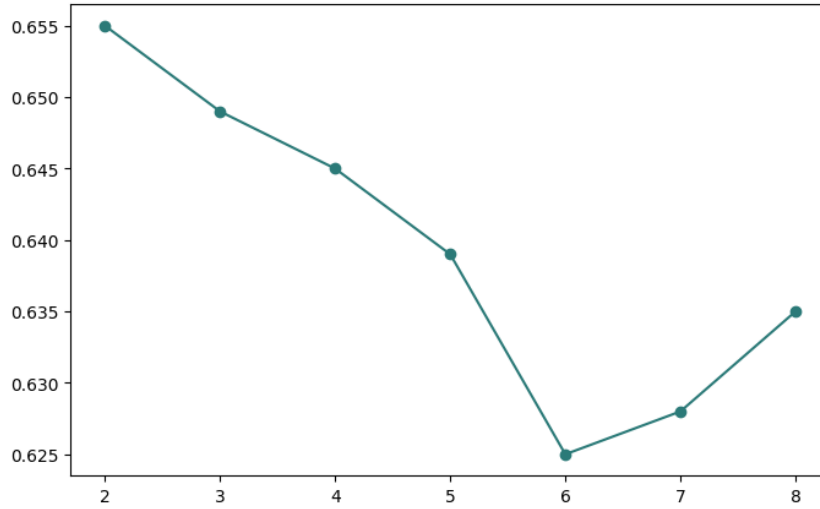


Figure 8.3: Test accuracy vs. register size (qubit count), Experiment VI qubit-count sweep, Pi4Angle encoding.

Accuracy is highest at the smallest register size (2 qubits, 0.655) and *decreases* roughly monotonically as register size grows to 6 qubits (0.625), ticking back up slightly at 7–8 qubits. This is consistent with narrower registers meaning more repeated, weight-shared applications of the same small circuit (more chunks of the 8-d feature vector, each seen by a lower-capacity 2-qubit circuit) rather than one wide, higher-capacity 8-qubit circuit seeing the whole vector at once — on a training set of this size (3,600 images), the narrower, weight-shared configuration may generalize better precisely because it has less capacity to overfit, mirroring the same over-capacity concern that motivated Experiment I/II’s own `FC_LAYERS` reduction.

A second discrepancy worth flagging. The 8-qubit point of this sweep re-trains the identical configuration (Pi4Angle, 8 qubits, same CNN embedding, same PCA/LDA features, same training recipe) already reported in Table 8.1, yet reports 0.635 here versus 0.519 there — a 0.116 absolute difference for what should be the same experiment. See Sec. 8.8 for discussion.

8.5 Quantum-vs-classical ablation

To isolate how much of the full pipeline’s accuracy is attributable to the quantum circuit itself versus the two classical scaffolding mechanisms layered around it (the ResNet-style residual re-upload, Sec. 3.3.1, and the classical skip connection that concatenates raw CNN-embedding dimensions directly onto the classifier input), the Pi4Angle model was retrained under four configurations, disabling each mechanism independently and jointly.

Table 8.3: Experiment VI ablation: quantum circuit vs. classical scaffolding, Pi4Angle encoding.

Configuration	Reupload scale	Skip dim	Test acc. (Δ vs. full)
full	0.3	8	0.635 (+0.000)
no_reupload	0.0	8	0.622 (−0.013)
no_skip	0.3	0	0.430 (−0.205)
quantum_only	0.0	0	0.453 (−0.182)

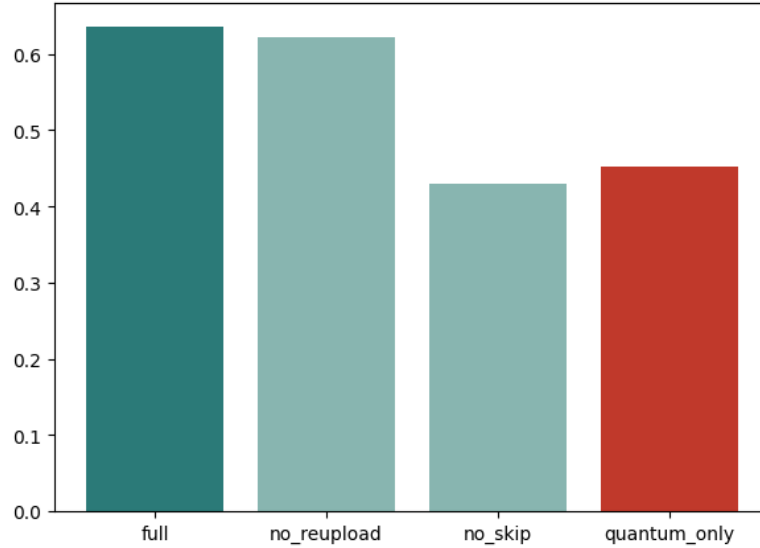


Figure 8.4: Ablation bar chart, Experiment VI: full model vs. removing the residual re-upload, removing the classical skip connection, and removing both (quantum-circuit-only). Red bar marks the quantum-only configuration.

Interpretation. Removing the residual re-upload alone costs very little (−0.013), suggesting the quantum-circuit-internal ResNet-style shortcut contributes only marginally to this model’s accuracy at this qubit budget. Removing the classical skip connection alone, however, costs −0.205 — by far the largest single effect in this ablation — and the quantum-only configuration (both mechanisms disabled) scores 0.453, only modestly above the 0.430 scored by keeping the residual re-upload but removing the skip connection. Taken together, these numbers indicate that a substantial share of the full pipeline’s 0.635 test accuracy is being carried by the 8 classically-bypassed CNN-embedding dimensions concatenated directly onto the classifier input, rather than by the quantum circuit’s own processing of the encoded features. This is consistent with, and quantifies for the first time in this report, the concern raised throughout Chapters 3–4 that the pipeline’s accuracy ceiling is shaped at least as much by the classical CNN/PCA/LDA/skip-connection stages as by the quantum classifier itself.

8.6 Widened FPN-QCNN encoding sweep

Following up on Experiment II’s single-encoding (Pi4Angle-only) FPN-QCNN result, v7 widens the multi-scale sweep to include AQHE-DualEnc and FRQI as well, reusing the Experiment II (v6.1) FPN-QCNN configuration unchanged (QUBITS_PER_LEVEL=3, FPN_BRIDGE_ROUNDS=2, explore-then-shrink learning-rate schedule, FPN_EPOCHS_Q=20). **This run did not complete:** the notebook’s own execution log for this section stops after only 4 of the budgeted 20 epochs on the first (Pi4Angle) of the three encodings queued, with no test accuracy recorded for any

of the three encodings. The accompanying diagnosis markdown (carried over verbatim from Experiment II) continues to describe and analyze the earlier v6/v6.1 numbers (0.620 single-scale vs. 0.543 FPN-QCNN) rather than any result from this run, since no new result was produced to update it with. Consequently, this report cannot draw any FPN-QCNN conclusion specific to v7's widened sweep; the Experiment II conclusions (Sec. 4.3) remain the most recent completed FPN-QCNN evidence in this BTP.

8.7 QCNN-vs-kernel-SVM rank correlation

The notebook includes a section designed to merge this chapter's per-encoding QCNN test accuracies with the Experiment V kernel-SVM per-encoding accuracies (Table 7.2) and compute a Spearman rank correlation between the two architectures' encoding preferences. This section expects the SVM results to be supplied as an external JSON file or a manually-populated dictionary; **neither was provided when this notebook was executed**, and the corresponding cell produced no output at all. No rank-correlation result is therefore available from this run. Carrying out this comparison using the Table 7.2 figures from Experiment V is left as future work (Sec. 10).

8.8 Assumptions, caveats, and notebook-state inconsistencies

In the interest of transparency, this chapter reports the following inconsistencies exactly as they appear in the executed notebook, rather than silently resolving them:

- **The same (Pi4Angle, 8-qubit) configuration produced two different test accuracies within the same notebook run:** 0.519 in the main benchmark (Sec. 8.3) and 0.635 in the qubit sweep's 8-qubit data point (Sec. 8.4), a 0.116 absolute gap on nominally identical settings.
- **Two plotting cells raised a `NameError`:** `name 'chance' is not defined` immediately after printing correct-looking numerical results (the qubit-sweep and ablation summary cells both print their tables successfully, then error while adding the chance-level reference line to their accompanying plot). A `NameError` on a variable that is assigned earlier in the same notebook (`chance = 1.0 / CONFIG["NUM_CLASSES"]`, defined during the main benchmark cell) is a signature of cells having been executed out of top-to-bottom order, or of an intervening cell re-running in a way that cleared or shadowed that global variable before the later cells ran — both symptoms consistent with (though not conclusive proof of) the same non-sequential-execution explanation offered for the accuracy discrepancy above.
- **The widened FPN-QCNN sweep (Sec. 8.6) did not run to completion, and the rank-correlation section (Sec. 8.7) was not run at all** (no external SVM results were supplied). Both are reported here as incomplete rather than populated with placeholder or estimated numbers.
- Given the above, the qubit-count sweep (Sec. 8.4) and ablation (Sec. 8.5) results — both of which are internally consistent with each other, sharing the same 0.635 full-model reference point — are judged the more reliable numbers from this chapter, while the main 10-encoding ranking (Table 8.1) should be treated as indicative of relative encoding ordering under one particular execution state rather than as a precise, reproduced-on-demand accuracy figure. Re-running the full notebook in a single, uninterrupted top-to-bottom pass is the direct way to confirm which set of numbers is representative, and is recorded as a priority item in Sec. 10.

8.9 Summary of observations (Experiment VI)

- Completing the full 10-encoding benchmark (rather than only the top-3 subset used in Experiments I–II) shows all ten encodings clustering within a narrow 0.024-wide band (0.495–0.519), with Pi4Angle — the simplest encoding in the benchmark — coming out on top under this run’s conditions, though see Sec. 8.8 for why this ranking should be treated cautiously.
- The qubit-count sweep (2–8 qubits) shows a mild *negative* trend in test accuracy as register width increases, consistent with narrower, weight-shared registers acting as an implicit capacity regularizer on this training-set size.
- The ablation is this chapter’s most load-bearing result: removing the classical skip connection costs far more accuracy (−0.205) than removing the quantum residual re-upload (−0.013), indicating the classical CNN-embedding bypass carries a disproportionate share of the full pipeline’s accuracy relative to the quantum circuit’s own contribution at this qubit budget.
- Two of the five planned v7 follow-ups (the widened FPN-QCNN sweep and the QCNN-vs-kernel-SVM rank correlation) did not produce usable results in this run and remain open items for a future, uninterrupted execution.

Chapter 9

Cross-Experiment Discussion

9.1 Consolidated results

Table 9.1 places the headline result from each experiment side by side. Note that these numbers are *not* directly comparable to one another, since they differ in dataset size, feature-extraction pipeline (trained CNN embedding vs. raw-pixel PCA/LDA), classifier type (custom QCNN vs. quantum-kernel SVM vs. VQC), and qubit budget; they are presented together only to summarize the BTP’s overall experimental trajectory.

Table 9.1: Cross-experiment summary of best-reported test accuracy.

Experiment	Best configuration	Test acc.	Dataset size
I — Single-scale hybrid CNN+QCNN	AQHE-DualEnc	0.620	5,000 images
I — FPN-QCNN (multi-scale, v1)	Pi4Angle	0.543	5,000 images
II — Single-scale hybrid CNN+QCNN (longer CNN budget)	AQHE-DualEnc	0.615	5,000 images
II — FPN-QCNN (multi-scale, post-fix)	Pi4Angle	0.465	5,000 images
III — Learnable quanvolution + QCNN (intensity/edge/texture, 1 layer)	Pi4Angle (all channels)	0.260	1,500 images
IV — Learnable quanvolution + QCNN (RGB, 2-layer fusion)	Pi4Angle (all channels)	0.217	1,500 images
V — Quantum-kernel SVM (18 encodings)	AQHE-DualEnc	0.320	600 images
V — PCE quantum kernel (refined, v2)	PCE (PCA-24, 2-layer)	0.273	600 images
V — VQC (gradient-trained)	FRQI	0.147	600 images
VI — Single-scale hybrid CNN+QCNN (full 10-enc. sweep)	Pi4Angle	0.519	5,000 images
VI — Same config, qubit-sweep re-run (8 qubits)	Pi4Angle	0.635	5,000 images
VI — Ablation: quantum-circuit-only (no skip/reupload)	Pi4Angle	0.453	5,000 images

9.2 Recurring themes across experiments

1. **Feature-extraction quality dominates encoding choice.** The single largest jump in accuracy across all four experiments came not from any change of quantum encoding or circuit architecture, but from replacing a raw-pixel PCA front end with a trained classical CNN embedding (Experiment I’s v2→v3 transition) and from scaling dataset size (400→5,000 images). AQHE-DualEnc’s accuracy nearly doubles between the 600-image raw-pixel-PCA/LDA pipeline of Experiment V (32.0%) and the 5,000-image trained-CNN pipeline of Experiments I–II (61.5–62.0%), despite using an *identical* quantum encoding circuit in both cases.
2. **Multi-scale/multi-register quantum fusion under-performed single-scale encoding at every qubit budget tested — and adding a second fusion layer made a quanvolutional pipeline worse, not better.** The FPN-QCNN variant (Experiments I–II) and the learnable quanvolutional feature extractor (Experiment III) both attempt to give the quantum circuit access to richer, spatially-structured information by splitting it across multiple registers or patches, and both under-performed a single, appropriately-dimensioned quantum register fed by a well-optimized classical embedding. Experiment IV pushes this pattern further: stacking a second, trainable fusion quanvolution layer on top of Experiment III’s single-layer pipeline (and switching to raw RGB input channels) *reduced* test accuracy from 26.0% to 21.7% despite adding trainable capacity and per-sample compute. In every diagnosed case across all four architectures, the root cause traced back to some form of information starvation or excessive compression *before* fusion, rather than to an inherent limitation of multi-register quantum processing itself — adding a fusion stage without first fixing the bottleneck it fuses through simply relocates and compounds the same problem.
3. **Optimization landscape issues (plateaus, aggressive LR decay) are as consequential as architectural choices.** Three of the four experiments (I, II, IV’s VQC run) explicitly document optimizer plateaus or unstable learning-rate schedules materially affecting final accuracy, independent of encoding or architecture quality.
4. **Quantum-kernel SVMs outperformed gradient-trained VQCs** at comparable qubit counts and sample sizes (Experiment V), consistent with the broader QML literature’s observation that variational circuit training is more susceptible to barren-plateau-type optimization difficulties than kernel-based approaches at small-to-moderate qubit counts [6, 4].
5. **AQHE-DualEnc is the most consistently strong encoding** across every experiment in which it was benchmarked, suggesting its hybrid amplitude-and-logarithmic-phase construction is a reasonable default choice for future BTP iterations — with the caveat that Experiment VI’s completed 10-encoding run placed it third rather than first, one of several signs (below) that not every result in this BTP should be treated as a precise, reproducible figure.
6. **Classical scaffolding can carry more of the final accuracy than the quantum circuit itself.** Experiment VI’s ablation is the only point in this BTP where the residual re-upload and classical skip connection were each independently disabled; removing the classical skip cost -0.205 (versus -0.013 for removing the quantum residual re-upload), indicating that for this architecture and qubit budget, the classically-bypassed CNN-embedding dimensions are doing more work than the quantum circuit’s own transformation of the encoded features. This directly quantifies a concern that was only qualitatively raised in Chapters 3–4.

7. Not every number in this BTP is reproducible within the same notebook run.

Experiment VI's main benchmark and its qubit-count sweep report two different test accuracies (0.519 vs. 0.635) for what should be an identical (Pi4Angle, 8-qubit) configuration, alongside `NameErrors` consistent with out-of-order cell execution. This is recorded transparently (Sec. 8.8) rather than papered over, and is a reminder that in an iterative notebook-driven BTP workflow, execution-order discipline is itself a methodological concern on par with the architectural ones this report otherwise focuses on.

Chapter 10

Conclusion and Future Work

10.1 Conclusion

This report has consolidated four experimental notebooks from a Bachelor’s Thesis Project on hybrid quantum–classical architectures for CIFAR-10 image classification. The strongest overall result — 62.0% test accuracy on a 5,000-image, 10-class balanced CIFAR-10 subsample, against a 10% chance level — was obtained by a single-scale hybrid pipeline combining a trained classical CNN feature extractor, PCA→shrinkage-LDA dimensionality reduction, an AQHE-DualEnc quantum data encoding, and a ResNet-style residual QCNN classifier with a classical skip connection. Two architectural extensions explored in this BTP — Feature-Pyramid-style multi-scale quantum fusion and fully learnable patch-based quanvolution — both under-performed this single-scale baseline at the qubit budgets and dataset sizes tractable under classical simulation, and this report has documented in detail the diagnosed root causes (information starvation relative to the single-scale baseline, shallow cross-register fusion, and unstable learning-rate schedules) together with the partial fixes attempted and their outcomes, including a negative result (Experiment II’s FPN-QCNN re-run performing worse than its own predecessor). A separate, large-scale benchmark of eighteen quantum data-encoding strategies confirmed AQHE-DualEnc as the strongest performer across a very different preprocessing pipeline and sample size, and showed that Pauli Correlation Encoding, despite favourable theoretical gate-count and capacity scaling relative to alternatives such as QAOA-style encodings, does not by itself outperform well-tuned supervised low-dimensional encodings at this problem scale, though a shallower, feature-selected refinement recovered a substantial fraction of the gap.

10.2 Future work

Building directly on the diagnoses recorded across all four experiments, promising directions for continued work include:

- Re-running the single-scale hybrid pipeline with a frozen, pretrained (e.g. ImageNet-pretrained `resnet18`) backbone in place of the from-scratch CNN, since several notebooks flag this as the most likely remaining lever for the single-scale ceiling observed across v3–v5;
- Systematic ablation of the residual re-upload scale and classical skip dimension (e.g. sweeping `RESIDUAL_REUPLOAD_SCALE` ∈ {0, 0.15, 0.3, 0.5} and `CLASSICAL_SKIP_DIM` ∈ {0, 4, 8, 16}) to isolate each residual pathway’s individual contribution;
- Deepening the *single-scale* circuit (more qubits, more `StronglyEntanglingLayers`) as an alternative to further multi-scale pyramid fusion, as suggested by Experiment II’s negative FPN-QCNN result;

- A matched-sample-size, matched-epoch-budget comparison between the classical-CNN hybrid pipeline and the learnable quanvolutional pipeline (Experiment III), and a controlled ablation freezing the quanvolutional weights to isolate the value of learnability itself;
- Extending the PCE encoding to $k = 3$ correlation order (capacity 1,512 at $n = 8$) to accommodate a larger PCA-128 feature set, and exploring PCE/other-encoding kernel ensembles ($K_{\text{final}} = \lambda K_{\text{PCE}} + (1 - \lambda) K_{\text{IQP}}$), as proposed in the notebook’s own next-steps list;
- A classical RBF-kernel SVM baseline on the same PCA-64 features used for PCE, to directly isolate whatever advantage (or disadvantage) the quantum kernel confers over a classical kernel on identical input features;
- Running the full benchmark suite on real quantum hardware or a noisy simulator to assess whether PCE’s substantially lower gate count translates into a practical robustness advantage under realistic noise, as its capacity/gate-count scaling would suggest;
- Re-executing `QCNN_CIFAR10_Hybrid_v7.ipynb` in a single, uninterrupted top-to-bottom pass to resolve the Sec. 8.8 discrepancy between its main benchmark and qubit-sweep results for the same configuration, and to complete the widened FPN-QCNN sweep and the QCNN-vs-kernel-SVM rank-correlation section, both of which did not produce results in the run reported in Chapter 8;
- Extending Experiment VI’s quantum-vs-classical ablation to every encoding (not only Pi4Angle) and to the FPN-QCNN and quanvolutional architectures, to check whether the classical-skip-dominance finding is specific to the single-scale Pi4Angle configuration or a general property of this BTP’s residual/skip design.

Bibliography

- [1] I. Cong, S. Choi, and M. D. Lukin, “Quantum convolutional neural networks,” *Nature Physics*, vol. 15, pp. 1273–1278, 2019.
- [2] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature pyramid networks for object detection,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2117–2125.
- [3] M. Henderson, S. Shakya, S. Pradhan, and T. Cook, “Quantum convolutional neural networks: powering image recognition with quantum circuits,” *Quantum Machine Intelligence*, vol. 2, no. 1, p. 2, 2020.
- [4] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, “Supervised learning with quantum-enhanced feature spaces,” *Nature*, vol. 567, pp. 209–212, 2019.
- [5] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Physical Review Letters*, vol. 122, no. 4, p. 040504, 2019.
- [6] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii, “Quantum circuit learning,” *Physical Review A*, vol. 98, no. 3, p. 032309, 2018.
- [7] M. Sciorilli, L. Borges, T. L. Patti, D. García-Martín, G. Camilo, A. Anandkumar, and L. Aolita, “Towards large-scale quantum optimization solvers with few qubits,” *Nature Communications*, vol. 16, p. 476, 2025.
- [8] P. Q. Le, F. Dong, and K. Hirota, “A flexible representation of quantum images for polynomial preparation, image compression, and processing operations,” *Quantum Information Processing*, vol. 10, no. 1, pp. 63–84, 2011.
- [9] Y. Zhang, K. Lu, Y. Gao, and M. Wang, “NEQR: a novel enhanced quantum representation of digital images,” *Quantum Information Processing*, vol. 12, no. 8, pp. 2833–2860, 2013.
- [10] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [11] A. Krizhevsky, “Learning multiple layers of features from tiny images,” Tech. Report, University of Toronto, 2009.
- [12] O. Ledoit and M. Wolf, “A well-conditioned estimator for large-dimensional covariance matrices,” *Journal of Multivariate Analysis*, vol. 88, no. 2, pp. 365–411, 2004.
- [13] V. Bergholm et al., “PennyLane: automatic differentiation of hybrid quantum-classical computations,” *arXiv preprint arXiv:1811.04968*, 2018.